UNIVERSITY COLLEGE MINI-PROJECT

HYBRID CNN-LSTM-PCA-SVM FOR FOUR-CLASS BRAIN TUMOR MRI CLASSIFICATION

GITHUB RESEARCH REPORT EDITION

Completed implementation and experimental investigation of five independent CNN backbones, 15 frozen hybrid pipelines, material duplicate leakage, external cohort overlap, domain shift and explainability.

Adapted from the authoritative revised 60-page final report. The original scientific chapters, 15 chapters of research content, appendices, tables, plots and substantially intact Streamlit screenshots are preserved. Identifying university submission front matter is omitted.

No project-member names or project-guide names appear in this edition. The project is described only as a university college mini-project.

Research-use warning: This is not a clinically validated diagnostic system, medical device, or lesion segmentation method. Its reported high original-split test scores were affected by material cross-split image duplicates.

Study completed in 2026 | GitHub documentation prepared 4 November 2026

About this edition

This PDF preserves the full research BODY of the revised university mini-project report: Chapters 1-15, references and Appendices A-F, corresponding to original report pages 1-49 (original physical PDF pages 12-60). Its preliminary cover, certificate, acknowledgements and other identifying submission pages are excluded; they are not research content. The original source report is not altered.

The publication front matter and repository navigation supplied here are new. Printed research page numbers inside the carried-over chapters retain their original numbering. They begin on PDF page 5 of this edition, so add four to a printed research-page number to find its physical PDF page.

The evidence hierarchy is: revised 60-page final report and current machine-readable GitHub records first; project-history notes for dated implementation decisions; the older 68-page report and historical GUI README only as superseded snapshots.

No results were retrained or newly evaluated to prepare this public edition. Existing numerical claims come from the project report and cited saved evidence. The historical QC report recorded 174/174 numerical checks and 75/75 local frozen-bundle hashes at the time it was generated. Current public GitHub does not include .pth neural checkpoints; full runtime reproduction from a fresh clone requires those original model files.

Both the original Nickparvar partition and the BDNeuro collection were audited for overlap. Internal original-split accuracy should not be presented as a clean independent estimate. Web-40 is a small, exploratory stress test, not a clinical validation cohort.

CHAPTER NAVIGATION

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Printed page numbering preserved from the revised report | PDF page = printed page + 4

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SOURCES AND EVIDENCE

Citation and evidence key

Development dataset

Masoud Nickparvar, Brain Tumor MRI Dataset, Kaggle. The study used 7,023 images (5,712 Training + 1,311 Testing). https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

Structured external dataset

Hira et al., BDNeuro-MRI: A Bangladeshi Clinical Brain Tumor MRI Dataset for Four-Class Deep Learning Classification, Mendeley Data Version 8, published 8 September 2026. DOI 10.17632/zwr4ntf94j.8. https://data.mendeley.com/datasets/zwr4ntf94j/8

Primary research paper

Berete et al. (2026), Hybrid CNN + LSTM + SVM model for advanced brain tumor classification. DOI 10.1007/s13721-025-00718-8. https://doi.org/10.1007/s13721-025-00718-8

Web-40 image-level provenance

40 source pages and figure panels are documented in the repository workbook docs/DL_TUMOR_CLASSIFICATION_FINAL_CLAUDE_PACKAGE_2026-10-03/manifests/Web_Stress_Test_40_Source_Lis
The web images do not share a single upstream publisher/license.

Canonical verification paths

phase2/external_validation/frozen_15_manifest.json;
phase2/external_validation/comparison_15_models_bdneuro_full/external_comparison_15.csv;
phase2/external_validation/comparison_15_models_web_stress_40/external_comparison_15.csv;
final_project_docs/duplicate_leakage_audit/duplicate_leakage_audit.json.

Project repository and navigation

https://github.com/bakayaro05/Tumor-Classification-DL-. See README.md, docs/REPOSITORY_DOCUMENTATION.md, docs/DATASETS_AND_REFERENCES.md and docs/PUBLICATION_NOTES.md after these files are added to the public repository.

CHAPTER 1
INTRODUCTION

1.1 MOTIVATION

Magnetic resonance imaging (MRI) is a principal modality for suspected brain tumours, and reading large numbers of scans is time-consuming and expert-dependent. Hybrid classifiers built on ImageNet-pretrained convolutional neural networks (CNNs) now report 98–99% accuracy on public four-class brain-MRI benchmarks. The task studied here is single-image classification of two-dimensional MRI slices into **glioma**, **meningioma**, **pituitary** tumour or **no tumour** (`notumor`); these are dataset-supplied tumour-type labels, not grades, and the input is a raster image, not a DICOM volume.

When reported accuracy is already near-perfect, the important question is **whether that score can be trusted**. A headline number can mislead in four ways: the training protocol may hide how models converge; identical images on both sides of a split (**data leakage**) can inflate the test score; performance may hold only on images resembling the training collection (**domain shift**); and correct predictions may rest on regions unrelated to the lesion (**model opacity**). This report investigates all four for one recent hybrid architecture.

1.2 RESEARCH QUESTIONS AND OBJECTIVES

The project began from the Review-I problem statement: *to build and rigorously evaluate a hybrid CNN–LSTM–PCA–SVM framework for four-class brain MRI classification, examining the effect of CNN backbone choice, training behaviour, data integrity and generalization — as a research framework, not a clinical diagnostic device.* As evidence accumulated, this sharpened into one central question: **can the hybrid classifier achieve strong four-class performance, and how trustworthy are those results once training behaviour, duplicate leakage, external generalization and explanation behaviour have been examined?** The question decomposes into seven research questions (Table 1.1). Several arose directly from the answer to the previous one, and the chapters follow that order.

Table 1.1 Research questions, the evidence that raised each one, and where it is answered.

RQ	Question	Why it arose	Phase	Ch.
1	Can the base-paper hybrid reproduce strong four-class performance across five backbones, and does PCA + SVM add value?	The base paper reports 94–99% within-dataset accuracy	1	4
2	Are the high internal scores trustworthy as evidence of generalization?	Near-99% accuracy assumes independent partitions, never verified	2D	5
3	Was early stopping hiding convergence behaviour?	Phase 1 backbones ran for unequal, early-stopped budgets	2A	6
4	Which training choices matter most, and do small tuning gains transfer?	The common baseline's sensitivity was unknown	2B	7, 10
5	Do internally selected, frozen models transfer to a different structured dataset?	Leakage meant internal evidence could not stand alone	2C	8
6	How do they behave under severe, uncontrolled domain shift?	A practical question a curated dataset cannot answer	2C+	9
7	Does correct classification imply lesion-centred reasoning?	Accuracy alone is silent about what the model uses	4	11

Phase labels link each question to the Review-I roadmap.

Chapter 10 (Phase 3) synthesizes the answers and names a final project representative; Chapter 12 (Phase 5) describes the research prototype. No accuracy target was set.

Scope. In scope: five backbones, RMSprop, the Nickparvar dataset for development, and BDNeuro-MRI V8 and a 40-image web set as evaluation-only data. Out of scope: the full optimizer grid, segmentation, DICOM/volumetric input, patient-level analysis (no patient identifiers exist) and any clinical claim. The data-integrity study (Phase 2D) is reported before Phase 2A because it was run first and changed how all later internal numbers are read. Complete tables (all 15 external pipelines, all 35 Phase 2B runs, per-class recall, per-case XAI results), tensor shapes, script and evidence inventories, the implementation history and a glossary are given in Appendices A–F at the end of this report.

Claim conventions used throughout this report. Internal* — a result on the original Nickparvar split, which contains known cross-split duplicates (Chapter 5): historical, not clean independent generalization. **material_clean** — post-hoc leakage-aware test subset (n = 1,061); it does not repair training or validation. **BDNeuro full** (n = 2,644) — primary structured external cohort; **BDNeuro balanced** (n = 1,648) — nested sensitivity subset, not a second dataset. **Web-40** — exploratory, source-verified stress test (one image = 2.5 pp), not clinical validation. **XAI maps** — coarse contribution/sensitivity maps, not segmentation. Patient-level independence cannot be established; one seed and one split; no statistical significance is claimed.

CHAPTER 2
RELATED WORK, BASE PAPER AND RESEARCH GAP

2.1 BASE PAPER

The methodological anchor is M. Bérété, A. Echtioui, L. Sellami and A. A. Ben Hmida, "Hybrid CNN+LSTM+SVM model for advanced brain tumor classification," *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, art. 43, Springer, 2026 [1]. It was chosen because it is recent, uses the same four-class dataset and defines five comparable backbones in one hybrid design: a CNN feature map, reshaped into a sequence, passes through an LSTM, batch normalization, dropout and dense layers, and the features are standardized, reduced by PCA to 64 components and classified by a linear SVM. The reported protocol uses four optimizers ($5 \times 4 = 20$ configurations), learning rate 1e-4, at most 30 epochs, batch size 32, dropout 0.3, and flip and colour augmentation; its reported results are in Table 2.1.

Table 2.1 Base-paper results under RMSprop, as reported by the authors [1, Table 3].

Model (RMSprop)	Accuracy	Precision	Recall	F1
VGG16 + LSTM + SVM	94%	94%	93.5%	93.5%
VGG19 + LSTM + SVM	96%	95.5%	95.25%	95.25%
ResNet50 + LSTM + SVM	99%	99%	98.75%	98.75%
ResNet152 + LSTM + SVM	99%	99%	99%	99%
EfficientNetB3 + LSTM + SVM	98%	98%	97.75%	98%

These are the paper's numbers, not this project's.

The paper reports no duplicate check on its partitions, no external dataset and no implemented explainability method (Grad-CAM and Score-CAM are listed as future work). Because it used the same public dataset and supplied split, its reported result should be interpreted as a within-dataset benchmark; this project did not independently reproduce or audit the authors' exact experimental run.

What this project inherited and what it added. The project kept the base paper's core flow and re-implemented it for the same five backbones (Fig. 2.1(b), upper box; deviations in Table 3.1). Everything in the lower part of Fig. 2.1(b) — the leakage audit, fixed-horizon and sensitivity studies, frozen 15-pipeline external evaluation with its own overlap audit, Web-40 stress test, formal explainability study and prototype — was **added by this project and is not part of the base paper**.

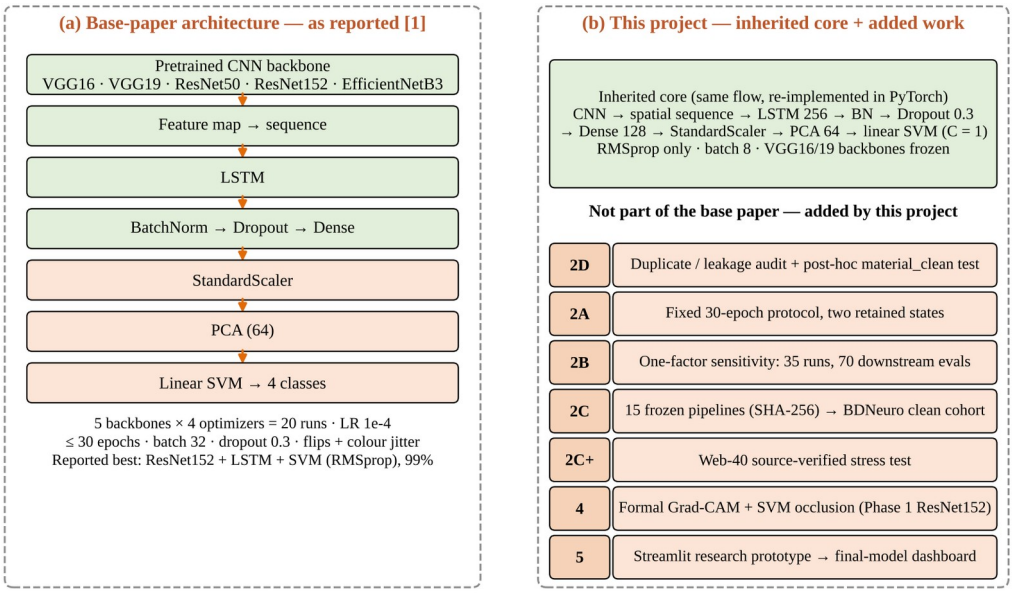


Fig. 2.1 Base paper versus this project.

(a) Base-paper architecture and protocol as reported by Bérété et al. [1] (flow as shown in the Review-I presentation). (b) This project: the inherited core, re-implemented, and the evaluation work added by this project (phase numbers in tags). Panel (b) components are not claimed for the base paper.

2.2 RELATED WORK AND RESEARCH GAPS

The Review-I survey covered hybrid CNN–recurrent classifiers [9], ensembles of fine-tuned networks [11, 13], residual-network classifiers [15], optimization-driven CNN design [14] and Grad-CAM explanation [10, 12]. Typically accuracy is reported on one public dataset, data independence is assumed, external validation is left as future work, and heatmaps are read as localization. Table 2.2 maps each gap to the question and chapter that address it; "(obs.)" marks this project's reading rather than the authors' statement.

Table 2.2 Gaps identified in the Review-I literature survey and the project's response.

Evidence in literature	Gap	Project response	RQ / Ch.
Bérété et al. [1]	No duplicate/leakage check	Cross-split duplicate audit; post-hoc leakage-aware test	2 / 5
[1], [9], [12]	Single public dataset; no external validation (obs.)	15 frozen, hashed pipelines on an overlap-audited external cohort	5 / 8
Review-I survey	Early stopping hides convergence	Fixed 30-epoch protocol, two states evaluated	3 / 6
Sahoo et al. [14]	Search-based tuning hides factor effects (obs.)	One-factor sensitivity study	4 / 7
[11], [13], [15]	Ensembles hide per-backbone behaviour (obs.)	Five independent pipelines, never combined	1 / 4, 10
[10], [12]	Grad-CAM cannot explain an external SVM; heatmaps over-read as localization (obs.)	Grad-CAM plus final-SVM occlusion, rated as coarse maps	7 / 11
(added during project)	Severe domain shift untested	Locked, source-verified Web-40 stress test	6 / 9

CHAPTER 3
MATERIALS AND METHODS

3.1 STUDY DESIGN

Fig. 3.1 summarizes the study as five layers, from separated data sources through integrity checks, model development and a freeze step to analysis and delivery. Two rules run through every layer: **evaluation data never influence fitting or candidate selection** (the retrospective naming of a representative, Section 10.4, is the one stated exception), and **every experimental number is traceable to a saved artifact** (Appendix E).

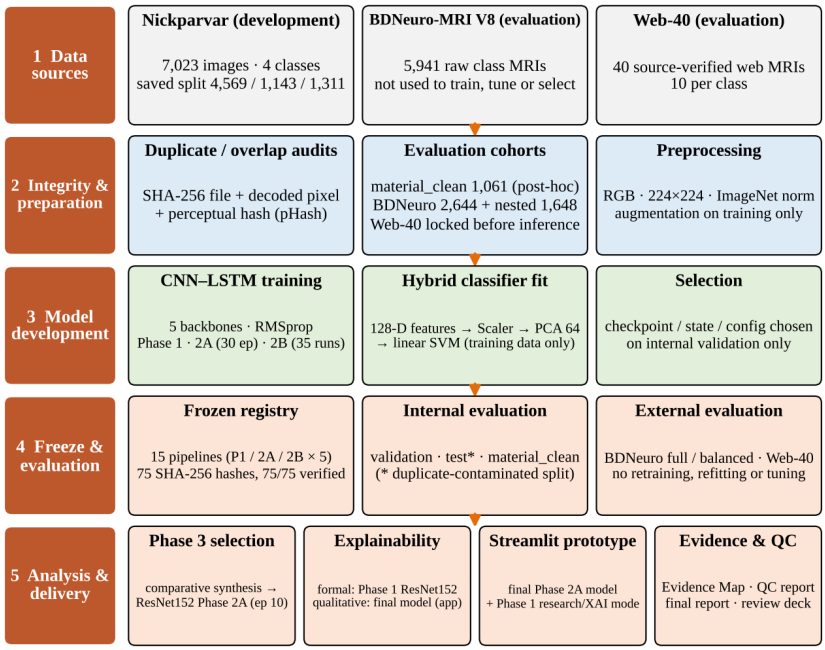


Fig. 3.1 High-level design of the complete study.

* internal test = known duplicate-contaminated split; material_clean = post-hoc leakage-aware subset; the balanced BDNeuro subset is nested in the 2,644-image cohort.

3.2 DATASETS

Development. The public **Brain Tumor MRI Dataset** by M. Nickparvar on Kaggle [16], the same four-class dataset as the base paper: 7,023 two-dimensional MRI images supplied as a *Training* folder (5,712) and a *Testing* folder (1,311): glioma 1,621 (1,321 / 300), meningioma 1,645 (1,339 / 306), notumor 2,000 (1,595 / 405) and pituitary 1,757 (1,457 / 300). Classes are reasonably balanced (largest-to-smallest ratio about 1.23 : 1). **External evaluation.** BDNeuro-MRI Version 8 [17], a structurally compatible four-class collection from a different documented source (Chapter 8), and Web-40, a source-verified set of 40 web MRI figures (Chapter 9). Neither was used for training, tuning or candidate selection.

3.3 IMPLEMENTED ARCHITECTURE

Each backbone defines one complete, independent pipeline (Fig. 3.2); the five pipelines are alternatives compared under a common protocol (with the VGG freezing exception of Section 3.4), **not an ensemble**. VGG16/VGG19 [2], ResNet50/ResNet152 [3] and EfficientNet-B3 [4] load ImageNet weights [18] and are truncated before global pooling, so a 224 × 224 input yields a 7 × 7 feature grid with 512, 2,048 or 1,536 channels. **The 7 × 7 map is reshaped into 49 spatial positions read in raster order by a single-layer LSTM** [5] (hidden size 256); the sequence therefore represents positions within one two-dimensional image, not time or MRI slices. Whether the LSTM adds anything beyond pooling was not tested. The last LSTM output passes through BatchNorm [19], Dropout 0.3 [20] and a Dense 128 layer with ReLU to give a **128-dimensional deep feature**.

Two branches read that feature. A temporary **neural head** (Linear 128 → 4) trains the network end to end and is used for comparison and Grad-CAM. The **final hybrid classifier** — StandardScaler → PCA reducing the 128-D representation to 64 components → linear SVM [6] (`SVC(kernel="linear", C=1.0, probability=False)`) — produces the official prediction. The scaler, PCA and SVM are fitted on training features only; the 64 components retained 99.18–99.70% of training-feature variance across Phase 1 and Phase 2A; neither PCA size nor C was tuned. The SVM yields decision values, not probabilities. The system was implemented in PyTorch [21] and scikit-learn [22]; tensor shapes are in Appendix E.

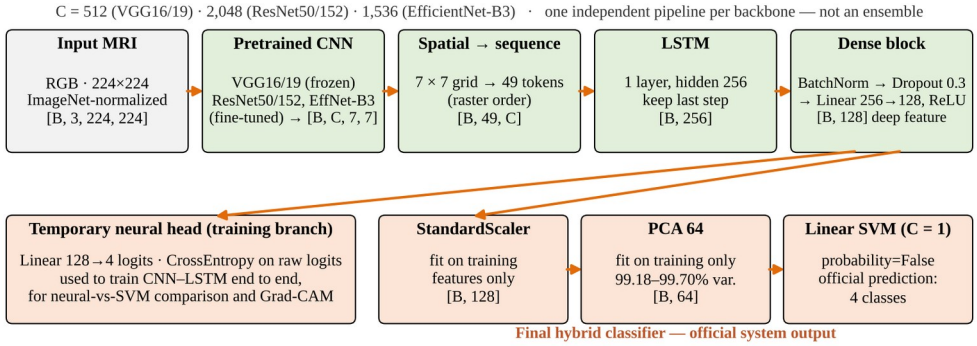


Fig. 3.2 Implemented architecture of one pipeline with tensor shapes.

Green: trained or adapted neural components; peach: the two downstream branches (temporary neural head and final hybrid classifier).

3.4 IMPLEMENTATION DECISION AFFECTING THE METHODOLOGY: FROZEN VGG BACKBONES

One implementation event changed how all cross-backbone results must be read. When fully fine-tuned under the common configuration, VGG16 collapsed: validation accuracy fell to about 28% (historical figure, no JSON preserved), the 128-D features became almost constant and PCA produced NaN values. notumor is 319 of 1,143 validation images (27.9%) — approximately the accuracy of always predicting the largest class. A diagnostic run showed healthy pretrained activations, placing the fault in the end-to-end training configuration; a single root cause was not isolated, and lower learning rates or partial

unfreezing were not studied. The VGG16 convolutional backbone was therefore frozen, with only the LSTM, BatchNorm, dense layer and head trained; VGG16 then trained stably and VGG19 received the same policy. ResNet50, ResNet152 and EfficientNet-B3 remained fine-tuned.

> **Consequence for interpretation.** The VGG networks act as fixed feature extractors with a trained read-out, whereas the other three adapt their convolutional weights. Cross-backbone differences in this report are therefore **differences between practical pipelines, not a pure architecture comparison**.

The remaining implementation history — an abandoned classical K-means/GLCM concept, a TensorFlow attempt that failed with `INVALID_PTX` on the RTX 5070 (sm_120) GPU and led to the move to PyTorch, and a first ResNet152 proof of concept — did not change the scientific design and is recorded in Appendix F.

3.5 PREPROCESSING, SPLIT AND TRAINING PROTOCOL

Every image is converted to RGB, resized directly to 224 × 224 and normalized with ImageNet channel statistics. No skull stripping, cropping or MRI-specific normalization is applied, so the model can see skull, background and borders. Augmentation is applied **only during training**: horizontal and vertical flips (p = 0.5 each) and `ColorJitter(brightness=0.1, contrast=0.2, saturation=0.2, hue=0.1)`, reproducing the base paper's ranges; all evaluation and prototype uploads use one deterministic transform.

The dataset supplies no validation set, so the Training folder was split 80/20 with stratification (seed 42) into **4,569 training** and **1,143 validation** images, with saved indices so that every experiment uses identical images; the 1,311-image Testing folder is the test set. Training data fit every fitted object, validation drives every checkpoint, state and candidate selection, and the test set is used only for final internal evaluation. All training used RMSprop, learning rate 1e-4, batch size 8 (with `drop_last`, because BatchNorm1d cannot use a one-image batch), seed 42 and best-validation checkpointing. As Chapter 5 shows, the supplied folders already shared duplicated content, so the split is **not** image-level independent. Table 3.1 lists every documented deviation from the base paper.

Table 3.1 Documented deviations from the base paper.

Element	Base paper	This project
Framework	TensorFlow	PyTorch (TensorFlow failed on the sm_120 GPU)
Optimizers	Adam, SGD, AdaGrad, RMSprop	RMSprop only
Batch size	32	8 (BatchNorm1d; drop_last)
Epochs	up to 30	Phase 1 early stopping (max 10 or 30); Phase 2A/2B fixed 30
LSTM / dense width	not fully specified	256 (128/512 in 2B) / 128 — implementation choices
Backbone freezing	not specified	VGG frozen after collapse; others fine-tuned
SVM probability	enabled	probability=False; relative scores
Augmentation	flips, brightness, contrast, saturation, hue	same effective ranges via ColorJitter
Added	—	leakage audit; frozen external validation; XAI; prototype

3.6 EVALUATION DESIGN

Each research question has its own evaluation and safeguard (Appendix E). Accuracy is reported with **macro-F1**, which weights every class equally. External results are also reported as a **gap**: external accuracy minus the same pipeline's internal full-hybrid validation accuracy.

CHAPTER 4
INTERNAL PERFORMANCE: CAN THE HYBRID ARCHITECTURE REPRODUCE STRONG RESULTS? (PHASE 1)

The first question was whether the re-implemented pipeline reaches within-dataset performance comparable to the base paper's 94–99% for all five backbones. Phase 1 produced **five independent pipelines, not an ensemble**. Each was trained with early stopping (patience 3) under unequal budgets: the frozen VGG16/VGG19 had 30 epochs and stopped at 19 and 21; ResNet50, ResNet152 and EfficientNet-B3 had 10 (10, 10 and 9 run). The scaler, PCA and SVM were fitted on training features, and the neural head and hybrid classifier were scored on the same checkpoints (Table 4.1).

Table 4.1 Phase 1: neural head versus PCA + linear SVM on the same checkpoints (val = internal validation).

Backbone	Best epoch	Epochs run / max	Neural head val	Neural head test*	SVM val	SVM test*	SVM test F1*	Δ test (SVM − head)
VGG16	16	19/30	94.58%	93.67%	94.93%	94.97%	94.68%	+1.30 pp
VGG19	18	21/30	94.58%	93.14%	94.49%	93.52%	93.10%	+0.38 pp
ResNet50	7	10/10	97.73%	97.33%	97.55%	98.09%	98.01%	+0.76 pp
ResNet152	8	10/10	98.25%	98.17%	98.25%	98.40%	98.32%	+0.23 pp
EfficientNet-B3	6	9/10	98.43%	98.02%	98.51%	98.47%	98.41%	+0.46 pp

* Original internal test = known duplicate-contaminated split (Chapter 5); never used for selection.

The hybrid pipeline worked for every backbone: internal* test accuracy ranged from 93.52% (VGG19) to 98.47% (EfficientNet-B3). The fine-tuned backbones sat about 3–5 pp above the frozen VGGs, reflecting architecture, freezing policy and budget together. The main confusion was glioma ↔ meningioma.

4.1 DOES THE DOWNSTREAM CLASSIFIER ADD VALUE?

Because the base paper's distinctive element is the PCA + SVM stage, the hybrid classifier was compared with the neural head it replaces. It raised test accuracy for all five backbones (+0.23 to +1.30 pp), but validation changes were mixed: +0.35 (VGG16), −0.09 (VGG19), −0.17 (ResNet50), +0.00 (ResNet152) and +0.09 pp (EfficientNet-B3). For ResNet152 the SVM corrected 11 neural-head test errors and introduced 8.

> **Interpretation and consequence.** RQ1 is answered positively within the dataset, though not at the base paper's exact 99%. The downstream classifier improved test but not validation accuracy consistently, so no universal benefit is claimed; because dimensionality reduction and classifier family change together, this is a **downstream-classifier comparison, not a PCA-only ablation**. The independence of the partitions had to be tested next (Chapter 5).

CHAPTER 5
DATA INTEGRITY: ARE THE HIGH INTERNAL RESULTS TRUSTWORTHY? (PHASE 2D)

By mid-September the project appeared ready to lock: five working pipelines at about 93.5–98.5% test accuracy, a prototype and explainability results. A pre-lock review found one unverified assumption beneath every result — that training, validation and test images were independent. If copies of an image sit on both sides of a split, the test set partly measures memorization (**duplicate leakage**). Phase 2D was planned at Review-I, but the audit was run first, on 20 September 2026, because its answer determined how every later internal number could be read.

5.1 AUDIT DESIGN

All three cross-split pairings (train ↔ validation, train ↔ test, validation ↔ test) of all 7,023 images were compared at three levels: SHA-256 of file bytes (byte-identical copies), SHA-256 of decoded pixels (identical images in different encodings) and perceptual hashing (pHash ≤ 4, dHash ≤ 6) confirmed by grayscale MAE and correlation. Strict confirmation made a pair a *high-confidence near-duplicate*; looser criteria (MAE ≤ 12, correlation ≥ 0.995) made it *review-only*.

5.2 FINDINGS

The audit returned **MATERIAL_DUPLICATION** (Table 5.1, Fig. 5.1): 562 material cross-split pairs — 464 exact pairs (304 exact-file + 160 exact-pixel-only) plus 98 high-confidence near-duplicate pairs — implicating 250 of the 1,311 test images. The 481 review-only perceptual candidates are **not** part of the 562.

Table 5.1 Cross-split duplicate audit of the original Nickparvar split.

Finding	Count
Exact file or exact decoded-pixel cross-split pairs	464 (304 file + 160 pixel-only)
Additional high-confidence near-duplicate pairs	98
Material cross-split pairs	**562**
Additional review-only perceptual candidates	481
Material pairs by pairing	train–test 344; train–val 146; val–test 72
Unique images in a material pair	train 354 (7.75%); val 122 (10.67%); **test 250 (19.07%)**
Implicated test images by class	glioma 0; meningioma 109; notumor 136; pituitary 5
Material pairs with different class labels	0

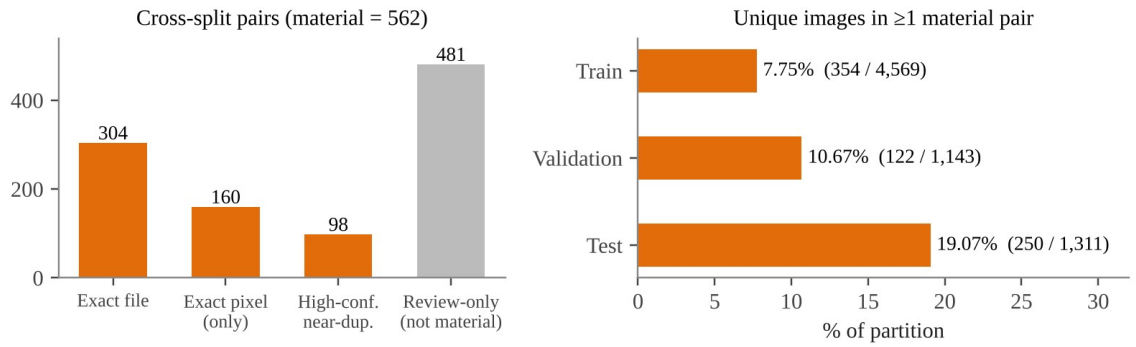


Fig. 5.1 Composition of the duplicate audit: pair types (left) and share of each partition involved in at least one material pair (right).

Pairs are not images. The 562 counts *pairs*; one image can belong to several pairs and a pair can link training to validation without touching the test set. The unique images involved are 354 training, 122 validation and 250 test images, so the leakage-aware test subset is 1,311 − 250 = 1,061 images; the number of affected images cannot be obtained by halving the pair count.

About one test image in five has a material counterpart in training or validation. Most of this overlap already existed between the dataset's supplied Training and Testing folders; the project's split did not create it. Every material pair carries the same label on both sides, so this is duplication, not label noise.

5.3 CONSEQUENCE: THE INTERNAL SCORES WERE REINTERPRETED

The audit did not invalidate the models, but it invalidated one interpretation: the statement that ~98% test accuracy measured independent generalization was withdrawn. The original scores remain valid historical measurements of the supplied split; they no longer represent clean independent generalization. The immediate post-audit record treated full duplicate-group-aware re-splitting and retraining as blocking and marked the project "NOT locked"; that record is preserved as the historical state, and its leakage warning remains valid. Project review then adopted the policy followed thereafter: keep original metrics, labelled as historical contaminated-split outputs; quantify a post-hoc leakage-aware internal test; continue Phase 2 with the caveat stated; seek stronger generalization evidence from an independently audited external dataset; and keep duplicate-group-aware retraining as future work, needed only for a clean internal-generalization claim. **Re-splitting and retraining were not performed.**

5.4 WERE THE HIGH SCORES EXPLAINED ONLY BY DUPLICATED TEST IMAGES?

The audit raised a second question: if the implicated test images were removed, would the high accuracy disappear? Each saved Phase 1 SVM was re-scored on its saved test features under three policies defined before re-scoring; nothing was retrained or refitted. **exact_clean** removes exact file/pixel duplicates (216 excluded, n = 1,095). **material_clean** — designated primary in advance, a post-hoc leakage-aware test subset — removes every

test image in a material pair (250 excluded, n = 1,061; class supports 300/197/269/295). **conservative_clean** also removes review-only candidates (385 excluded, n = 926; notumor falls from 405 to 134, so it is a robustness check rather than a replacement estimate).

Table 5.2 Phase 1 PCA + SVM test accuracy (%) under the post-hoc leakage policies.

Backbone	Original* (1,311)	exact_clean (1,095)	material_clean acc (1,061)	material_clean F1	Δ (pp)	conservative (926)
VGG16	94.966	94.977	94.816	94.302	-0.149	94.060
VGG19	93.516	93.242	93.025	92.415	-0.491	92.009
ResNet50	98.093	98.356	98.303	98.134	+0.210	98.056
ResNet152	98.398	98.447	98.398	98.315	-0.000	98.164
EfficientNet-B3	98.474	98.630	98.680	98.511	+0.206	98.488

Δ = material_clean − original. * Original internal test = known duplicate-contaminated split (Chapter 5); never used for selection. material_clean = post-hoc leakage-aware subset.

Interpretation and consequence. Removing the implicated test images changed accuracy by less than 0.5 pp for every backbone, and the fine-tuned-versus-VGG ordering was preserved. The high scores were therefore **not explained solely by the identified duplicated test images**. This does **not** show that the models are leakage-free: training and checkpoint selection used a split that itself contains duplicated content, so validation accuracy and best-epoch choices may be optimistic. material_clean removes contaminated *test* images; it does not repair contaminated *training*. The audit is image-level only (no patient identifiers), and near-duplicates below threshold may remain. RQ2 is therefore answered: the internal scores are informative but cannot, on their own, be read as independent generalization. This made an externally audited evaluation (Chapter 8) the decisive remaining experiment; the training-behaviour questions (Chapters 6–7) continued with this caveat stated.

CHAPTER 6

TRAINING BEHAVIOUR: WAS EARLY STOPPING HIDING CONVERGENCE? (PHASE 2A)

Phase 1 budgets differed (9–21 epochs run), so differences between backbones could reflect when training stopped rather than what each pipeline could learn. Phase 2A removed this confound by training all five baseline configurations (RMSprop, LR 1e-4, LSTM 256, dropout 0.3, Dense 128, batch 8, seed 42) for **exactly 30 epochs** with no early termination; all completed 30/30.

Two checkpoints were saved per backbone — `best_val_within_30` (highest neural validation accuracy within 30 epochs) and `epoch_30` — and **both** were passed through the full scaler → PCA → SVM path (10 evaluations), because the best neural checkpoint need not give the best SVM. The representative state of each backbone was chosen by a fixed rule: highest full-hybrid **validation** accuracy, then validation macro-F1, then — on an exact tie — preference for best_val_within_30. The test set played no part. The selected representatives were **VGG16 epoch 30, VGG19 epoch 30, ResNet50 epoch 30, ResNet152 epoch 10 and EfficientNet-B3 epoch 27, all in the best_val_within_30 state**. For VGG16, VGG19 and ResNet50 the best neural epoch was epoch 30 itself, so their two states give identical metrics and the tie rule selects best_val_within_30 (formally best_val_within_30 at epoch 30, not `epoch_30_model.pth`); full detail is in Appendix B.

Table 6.1 Phase 2A representative states versus Phase 1 (full-hybrid SVM validation accuracy).

Backbone	Best neural epoch	Phase 1 SVM val	Phase 2A SVM val	Phase 2A test*	material_ clean
VGG16	30	94.93%	95.888%	96.186%	95.853%
VGG19	30	94.49%	95.626%	95.271%	95.099%
ResNet50	30	97.55%	98.688%	98.932%	98.963%
ResNet152	10	98.25%	98.600%	98.703%	98.869%
EfficientNet-B3	27	98.51%	99.300%	99.314%	99.246%

Full Phase 2A detail in Appendix B. * Original internal test = known duplicate-contaminated split (Chapter 5); never used for selection. material_clean = post-hoc leakage-aware subset (n = 1,061).

Interpretation and consequence. Under the common 30-epoch protocol, every backbone's selected full-hybrid validation result exceeded its Phase 1 result; the ranking was EfficientNet-B3 (99.300%), ResNet50 (98.688%), ResNet152 (98.600%), VGG16 (95.888%), VGG19 (95.626%). The VGGs kept improving to epoch 30 (Phase 1 best epochs 16–18), so Phase 1 early stopping had halted them while they were still improving. ResNet152 peaked early (epoch 10) and EfficientNet-B3 late (epoch 27). Differences among the top three are fractions of a point on one seed and a contaminated split and are not claimed to be significant; on material_clean the seven distinct states moved by less than 0.35 pp. RQ3 is answered for the VGGs, and the 30-epoch protocol became the baseline for Phase 2B.

CHAPTER 7

MODEL SENSITIVITY: WHICH TRAINING CHOICES MATTER MOST? (PHASE 2B)

With a common 30-epoch baseline established, the next question was how sensitive the architecture is to ordinary training choices. If small changes in learning rate, LSTM width or dropout moved accuracy substantially, internal rankings would be fragile. A **one-factor-at-a-time** design around the Phase 2A baseline changed exactly one factor per run: learning rate (1e-5, 5e-5, 5e-4), LSTM hidden size (128, 512) or dropout (0.2, 0.5). Search-based tuning would find a good configuration but obscure each factor's effect; the aim was interpretable sensitivity. Five backbones × seven configurations gave **35 training runs, each trained for the full 30 epochs** (all completed 30/30) and, with both retained states evaluated downstream, **70 PCA + SVM evaluations**.

The test set was deliberately not evaluated. Phase 2B compares 70 states and picks the best per backbone; scoring the test set for each would turn it into a tuning set. Only training and validation features were extracted, and all Phase 2B selection is validation-based. Fig. 7.1 shows peak neural validation for every run.

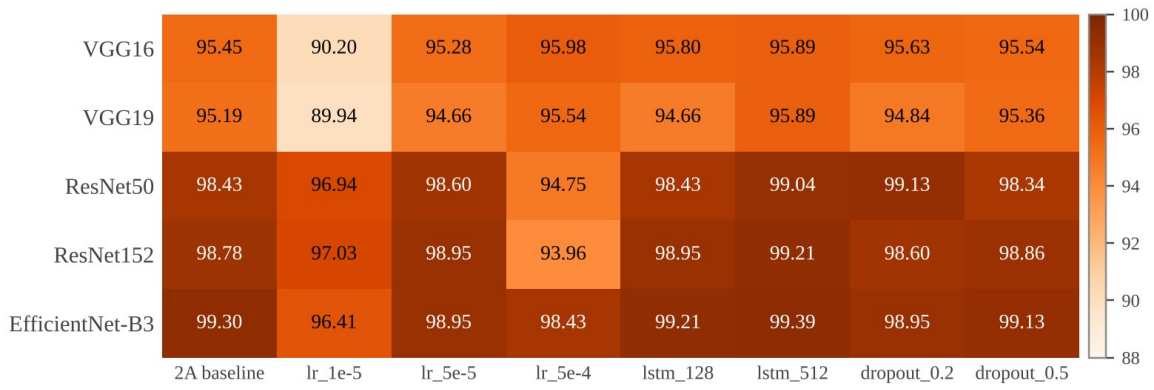


Fig. 7.1 Peak neural validation accuracy (%) for the Phase 2A baseline and each Phase 2B configuration (all 35 runs; full table in Appendix B).

Table 7.1 Within-backbone sensitivity ranges (pp) of peak neural validation accuracy, including the Phase 2A baseline, and full-hybrid validation of the selected Phase 2B representative (highest among 14 states; state detail in Appendix B).

Backbone	LR range	LSTM range	Dropout range	Phase 2B representative	Phase 2B SVM val	Phase 2A SVM val
VGG16	5.77	0.44	0.17	lr_5e-4 (ep 27)	96.500%	95.888%
VGG19	5.60	1.22	0.52	dropout_0.5 (ep 30)	95.538%	95.626%
ResNet50	3.85	0.61	0.79	dropout_0.2 (ep 28)	99.125%	98.688%
ResNet152	4.99	0.44	0.26	lstm_512 (ep 26)	99.300%	98.600%
EfficientNet-B3	2.89	0.17	0.35	lstm_512 (ep 29)	99.388%	99.300%
Mean	**4.62**	**0.58**	**0.42**			

- **Learning rate dominated, and its preferred value depended on the backbone.** LR 1e-5 underfitted the frozen VGGs (VGG16 90.20%, VGG19 89.94% peak), which tolerated 5e-4 well (VGG16 95.98%); for the fine-tuned ResNets 5e-4 was damaging (ResNet50 94.75%, ResNet152 93.96%). ResNet50 at LR 5e-4 showed the clearest late-epoch drift: 94.75% at epoch 25 to 90.99% at epoch 30 (−3.76 pp).
- **LSTM 512** raised peak neural validation for every backbone, but only by +0.09 to +0.70 pp, and several peaks were not maintained to epoch 30 — a modest peak-improvement association, not universal superiority.
- **Dropout** produced the smallest and least consistent changes; ResNet50 with dropout 0.2 reached its highest peak (99.13%).
- **Retained states:** across the 35 runs the best-neural state gave the higher SVM validation in 20 cases, the epoch-30 state in 8, and they tied in 7.

Interpretation and consequence. Learning rate was the dominant tested sensitivity: its mean range (4.62 pp) was about eight times that of LSTM width (0.58 pp) and eleven times that of dropout (0.42 pp). The selected Phase 2B representatives raised internal validation over Phase 2A for four of five backbones by 0.09–0.70 pp (VGG19: −0.09 pp). With one seed, one contaminated validation split and no interaction terms, these are controlled sensitivity findings, **not statistically established superiority claims**. Whether such small internal gains reflect better models or adaptation to one validation split cannot be answered internally — so both the Phase 2A and the Phase 2B representatives were frozen for external evaluation, so that the external evaluation could also check whether internal optimization carried over (Chapter 10).

CHAPTER 8
EXTERNAL GENERALIZATION: DO INTERNALLY STRONG MODELS TRANSFER? (PHASE 2C)

After the audit, internal numbers could not stand alone as generalization evidence, and even duplicate-group-aware retraining would not test transfer to a different source. External validation was therefore the most informative remaining experiment, and it had to be designed so that its results could not influence the models being tested.

8.1 FROZEN 15-PIPELINE PROTOCOL

For each backbone three full hybrid pipelines were taken forward — Phase 1, the Phase 2A representative and the Phase 2B representative — giving 15, so that the external test could also compare training protocols. **The 15 pipelines were selected using internal validation only, frozen and SHA-256 hashed before external performance evaluation.** The freeze manifest records the rule: *"Use INTERNAL full-hybrid validation accuracy; break ties by macro-F1; on an exact tie prefer best_val_within_30 over epoch_30."* It stores five hashes per pipeline (checkpoint, scaler, PCA, SVM, internal results), 75 in total, and is time-stamped 2 October 2026, 16:50 UTC, before the first external result file (3 October, 07:54 UTC); all 75 hashes were re-verified against the evaluated artifacts (75/75 match). No model was retrained, refitted, recalibrated or re-selected afterwards (Fig. 8.1).



Fig. 8.1 Freeze-and-hash protocol for the 15 external-evaluation pipelines.

8.2 IS THE EXTERNAL DATASET ACTUALLY INDEPENDENT?

The external dataset is **BDNeuro-MRI Version 8** [17], with the same four classes: 5,941 class MRIs (glioma 2,191, meningioma 1,478, notumor 787, pituitary 1,485). It documented a different source ("Epic & CSCR Hospital"), and its README reports within-dataset de-duplication but leaves licence and citation fields unfilled. Documented provenance is a claim, not evidence of independence, so before any model was evaluated the same audit used in Chapter 5 was run between all 5,941 BDNeuro and all 7,023 development images (Table 8.1).

Table 8.1 BDNeuro ↔ Nickparvar overlap audit (all 5,941 vs all 7,023 images).

Measure	Cross-dataset	Within BDNeuro
Exact file / exact decoded-pixel duplicate pairs	3,421 / 3,519	0 / 0
Unique BDNeuro images with exact-file / exact-pixel overlap	3,295 / **3,297**	—
Exact-pixel pairs with differing class label	0	—
pHash ≤ 6 / ≤ 4 candidate pairs	9,981 / 6,664	1,588 / 65

More than half of raw BDNeuro (3,297 of 5,941, 55.5%) was pixel-identical to development images, always with the same label. Evaluating on these would partly measure memorization; the project does not speculate about which lineage caused the reuse. The adopted rule (Fig. 8.2) was to **exclude exact decoded-pixel matches only**: brain MRI is anatomically repetitive, and slices from different patients at the same level can have similar perceptual hashes, so non-identical pHash candidates were treated as screening evidence, not proof, and retained.

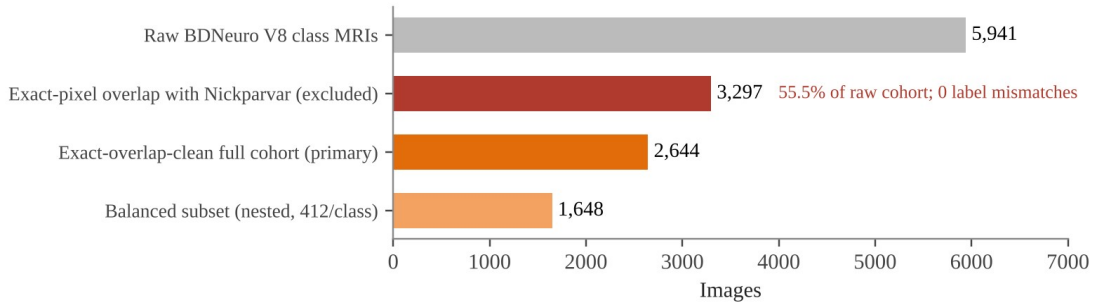


Fig. 8.2 From raw BDNeuro to the exact-overlap-clean full cohort and the nested balanced subset.

The **primary external cohort** is the 2,644-image exact-overlap-clean set (glioma 1,159, meningioma 472, notumor 412, pituitary 601), merged across BDNeuro's original splits because none was used for training or selection. Because it is imbalanced (glioma 43.8%), a deterministic **balanced sensitivity subset** of 412 per class (1,648) was drawn by SHA-256 ranking without model outputs. It is **nested** in the full cohort — a reweighting of the same images, not a second dataset — and on it accuracy equals mean per-class recall.

8.3 HOW WELL DID THE FROZEN MODELS TRANSFER?

All 15 frozen pipelines were evaluated with the deterministic transform (Fig. 8.3; macro-F1 and gaps in Appendix A). Every pipeline lost accuracy relative to its internal validation: accuracy ranged from **75.00% to 88.31%** on the exact-overlap-clean BDNeuro cohort, 10.29–20.54 percentage points below the corresponding internal validation values (gaps for every pipeline in Appendix A), and from 77.00% to 90.47% on the balanced subset. **ResNet152 Phase 2A** was strongest on both (88.31%, macro-F1 87.85% full; 90.47%,

macro-F1 90.50% balanced) with the smallest gaps (−10.29 and −8.13 pp). On average Phase 2A (83.21% full) exceeded Phase 2B (82.34%) and Phase 1 (79.77%).

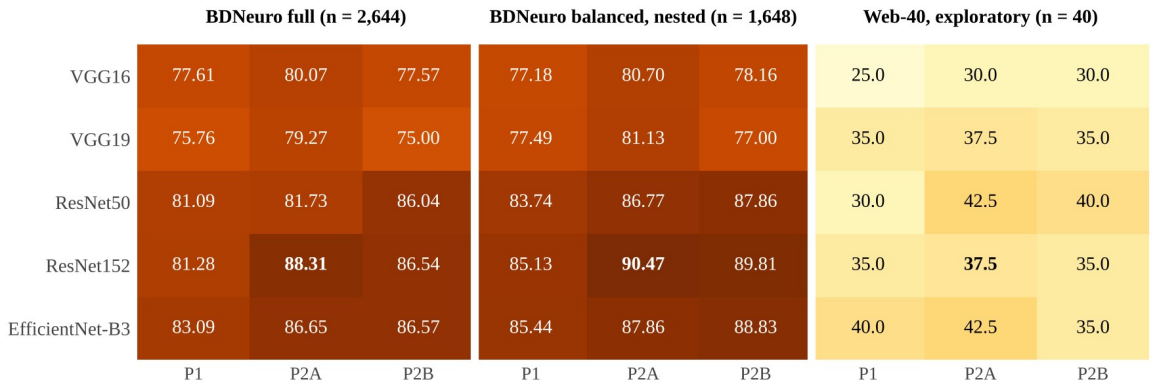


Fig. 8.3 External accuracy (%) of the 15 frozen pipelines by backbone and stage (P1 = Phase 1, P2A = Phase 2A, P2B = Phase 2B).

BDNeuro balanced is a nested sensitivity subset; Web-40 is an exploratory stress test (one image = 2.5 pp). Bold: final representative.

- **Higher internal validation did not predict higher external accuracy.** EfficientNet-B3 Phase 2B had the highest internal validation of all 15 (99.39%) but was not best externally; ResNet152 Phase 2A had the lowest internal validation (98.60%) of the six ResNet/EfficientNet Phase 2A/2B candidates yet was best externally.
- **Phase 2B tuning did not transfer consistently.** Phase 2B beat Phase 2A externally only for ResNet50 (dropout 0.2: 86.04% vs 81.73% full) and, on the balanced subset, EfficientNet-B3; the VGG Phase 2B representatives fell below Phase 2A (VGG19: 75.00% vs 79.27%).
- **Per-class transfer.** Pituitary had the lowest full-cohort recall in 12 of 15 pipelines and glioma in the other 3. Glioma recall was below the pipeline's mean class recall in 12 of 15, and accuracy on the balanced subset was higher in 14 of 15 pipelines (exception: VGG16 Phase 1, 77.61% full vs 77.18% balanced; per-class recalls in Appendix C).

Interpretation and consequence. RQ5 is answered **partially**: accuracy ranged from 75.00% to 88.31% on the exact-overlap-clean BDNeuro cohort, 10.29–20.54 percentage points below the corresponding internal validation values, and the internal ranking did not predict the external one. The overlap audit was itself a finding: a dataset documented as a different source contained 3,297 pixel-identical development images. BDNeuro is still a curated four-class collection of similar format, so it represents moderate shift; the next question was how the same frozen pipelines behave when that structure is removed.

CHAPTER 9

ROBUSTNESS UNDER SEVERE DOMAIN SHIFT: WHAT HAPPENS ON UNFAMILIAR WEB IMAGES?

The project guide asked a practical question that no curated dataset can answer: *what happens if a user uploads an MRI found on the web?* Publication and open-access figures differ from the training data in scanner, sequence, orientation, crop and resolution at once. Web-40 is a **source-verified exploratory stress test** designed to observe failure modes under that severe shift; it is not clinical validation or a population-level accuracy estimate, and its results were not used to train, tune or select any model.

Exactly 40 images, 10 per class, were collected, so accuracy equals mean recall and **one image = 2.5 pp**. Search engines were used only to *find* candidates; each diagnosis came from the original source page (case reports, open-access articles, medical reference sites) and was recorded with URL, figure panel, sequence and orientation. Only the indicated MRI panel was used, cropped without altering anatomy or contrast; CT, histology, segmentation maps, ambiguous diagnoses and heavily annotated figures were rejected. The set was audited against itself, all 7,023 Nickparvar images and the 2,644 BDNeuro images (exact file, exact pixel, pHash); two perceptual candidates required visual review (Table 9.1).

Table 9.1 Web-40 pre-inference review decisions.

Image	Evidence	Visual review	Decision
NOR_01	pHash 0 vs Nickparvar Te-no_0227 (pixels not identical; historical record, no JSON preserved)	Same axial T2 slice, different representation	**Rejected and replaced before inference**
MEN_02	pHash 6 vs Nickparvar Tr-me_0633	Clearly different scan	Retained

The re-run audit found 0 exact file or pixel overlaps and no pHash ≤ 4 matches. The cohort was then locked (`LOCKED_BEFORE_INFERENCE`, 10:37:22 UTC on 3 October; 15-model comparison output 10:38:09 UTC) with the rule that no image may be replaced, recropped or edited after predictions are viewed. The 15 frozen pipelines then produced **600 predictions**.

Accuracy fell to **25.0%–42.5%** (Fig. 8.3); stage means were 33.00% (Phase 1), 38.00% (Phase 2A) and 35.00% (Phase 2B). The highest values were ResNet50 Phase 2A and EfficientNet-B3 Phase 2A (both 42.5%, 17/40; macro-F1 38.18% and 31.97%); EfficientNet-B3 Phase 1 and ResNet50 Phase 2B reached 40.0%. Differences of one or two images cannot support a ranking.

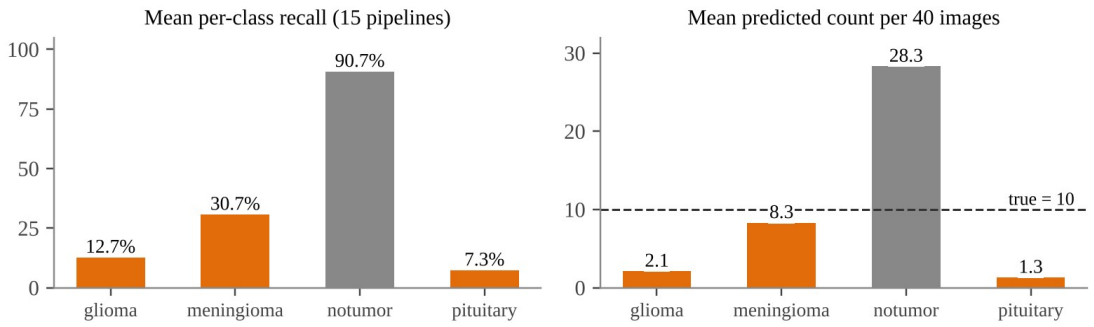


Fig. 9.1 Web-40 (exploratory): mean per-class recall and mean predicted count per 40 images across the 15 frozen pipelines.

Interpretation and consequence — no-tumour prediction bias. The failure was not random (Fig. 9.1). On average the models predicted `notumor` for about 28.3 of 40 images, although only 10 were normal; mean recall was 90.7% for notumor but 12.7% for glioma and 7.3% for pituitary. Under strong domain shift the classifiers systematically defaulted to "no tumour" — a particularly concerning false-negative pattern in a hypothetical screening context. Because scanner, sequence, slice level, orientation, cropping and resolution all change at once, the results are consistent with partial reliance on dataset-specific appearance cues, but without acquisition metadata the drop cannot be attributed to a single cause. RQ6 is answered for this exploratory set: the pipelines were not robust to severe presentation shift. Limits: n = 40 convenience sample, publication figures over-represent striking cases, diagnoses were taken from source pages without radiologist re-reading, and 42.5% is not "clinical accuracy".

CHAPTER 10

SYNTHESIS: DOES INTERNAL OPTIMIZATION PREDICT EXTERNAL ROBUSTNESS? (PHASE 3)

10.1 STAGE COMPARISON

Table 10.1 Stage means over the five backbones on the three external evaluations (accuracy; macro-F1 in parentheses).

Stage	BDNeuro full (2,644)	BDNeuro balanced (nested)	Web-40 (exploratory)
Phase 1	79.77% (78.60%)	81.80% (81.80%)	33.00% (26.56%)
Phase 2A	83.21% (82.25%)	85.39% (85.41%)	38.00% (30.56%)
Phase 2B	82.34% (81.26%)	84.33% (84.33%)	35.00% (26.95%)

The ordering **Phase 2A > Phase 2B > Phase 1** held on all three evaluations. Phase 2A (fixed 30-epoch horizon) was associated with stronger stage-mean performance than Phase 1 (early stopping) across the evaluated cohorts; this is an association, not an isolated causal effect. The differences are modest (Phase 2A − Phase 2B: 0.86 pp full, 1.06 pp balanced, 3.0 pp web, i.e. 1.2 images) and rest on five backbone pairs from one seed. On both BDNeuro cohorts every fine-tuned pipeline was stronger than every frozen-VGG pipeline (not so on Web-40); no backbone was best on every layer of evidence.

10.2 THE GENERALIZATION GRADIENT

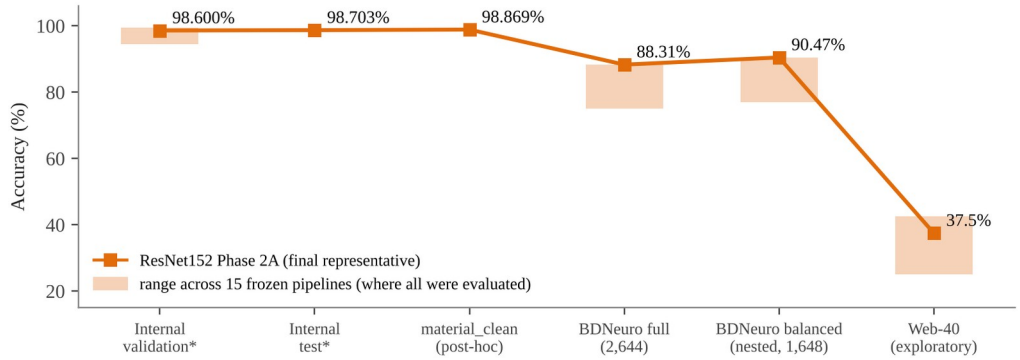


Fig. 10.1 Generalization gradient: the final representative across evaluation layers, with the range across all 15 frozen pipelines where all were evaluated.

* internal = known duplicate-contaminated split.

As evaluation moves from the training distribution (internal*, about 93–99%) through a structured but different source (BDNeuro full, 75.0–88.3%) to web images (25.0–42.5%), performance falls step by step for all 15 pipelines (Fig. 10.1).

10.3 WHY INTERNAL TUNING GAINS DID NOT CONSISTENTLY TRANSFER

The Phase 2B gains (0.09–0.70 pp, i.e. 1–8 of 1,143 validation images) were the maximum of 14 states per backbone on one validation split that overlaps with training. **The tuned Phase 2B gains did not consistently transfer externally. Selection optimism and domain shift are plausible contributors, but the experiments do not isolate a single causal explanation**; ResNet50 with dropout 0.2 shows that tuning *can* help but cannot be assumed to.

10.4 FINAL PROJECT REPRESENTATIVE

ResNet152 — Phase 2A standardized 30-epoch pipeline — best_val_within_30 — epoch 10. Configuration: RMSprop, LR 1e-4, batch 8, LSTM 256, dropout 0.3, Dense 128, StandardScaler → PCA 64 → linear SVM (C = 1). The choice had two steps. (1) The 15 pipelines were selected using internal validation only, frozen and SHA-256 hashed before external performance evaluation. (2) Naming one of them the representative was a retrospective synthesis of internal validation and the structured BDNeuro evidence. **Web-40 was not used** for this choice, and no model was altered after any external result.

Table 10.2 Evidence trajectory of the final representative (ResNet152 Phase 2A, best_val_within_30, epoch 10).

Evaluation layer	Accuracy (macro-F1)	Status
Internal full-hybrid validation*	98.600%	Used for candidate selection
Internal test, 1,311 images*	98.703%	Never used for selection
material_clean (n = 1,061)	98.869% (98.788%)	Post-hoc leakage-aware subset
BDNeuro full (n = 2,644)	88.31% (87.85%)	Primary external; gap −10.29 pp
BDNeuro balanced (n = 1,648)	90.47% (90.50%)	Nested subset; gap −8.13 pp
Web-40 (n = 40)	37.5% (31.66%)	Exploratory; not used for selection

* known duplicate-contaminated original split.

Its BDNeuro recall was lowest for glioma (83.5%; meningioma 94.9%, notumor 93.2%, pituitary 89.0%); on Web-40 it recognized 9 of 10 normal images but only 1 of 10 gliomas and 1 of 10 pituitary tumours. Because BDNeuro results informed step (2), its BDNeuro figures carry some **selection optimism** if read as the chosen model's performance; the 15-pipeline results (Appendix A) are the selection-independent record.

Why ResNet152 Phase 2A. The pipeline was compelling because it combined:

- strong internal full-hybrid validation (98.60%*) and 98.869% on the post-hoc material_clean subset;
- the **highest BDNeuro full-cohort accuracy** of the 15 frozen pipelines (88.31%) and the **highest BDNeuro balanced accuracy** (90.47%);
- the **smallest internal-to-external drop** (−10.29 pp on the full cohort; −8.13 pp balanced);

- these results with the simple Phase 2A baseline configuration, without the additional Phase 2B tuning.

Meanwhile, EfficientNet-B3 Phase 2B had higher internal validation (99.39%, the highest of all 15), and ResNet50 Phase 2A and EfficientNet-B3 Phase 2A did better on Web-40 (42.5% each), where ResNet152 Phase 2A reached only 37.5%.

ResNet152 Phase 2A was therefore **not universally superior**. It was selected because it gave the strongest and most consistent structured external transfer trajectory among the frozen candidates. It remains a research and educational prototype.

CHAPTER 11

EXPLAINABILITY: DOES CORRECT CLASSIFICATION IMPLY LESION-CENTRED REASONING? (PHASE 4)

Accuracy is silent about what the model uses to reach its decision, so the study asked: **does high classification accuracy mean that the model reaches its prediction by focusing on the visible lesion?** The base paper lists Grad-CAM only as future work. The formal, rated study used the **Phase 1 ResNet152 pipeline**, was completed on 20 September 2026, before Phase 2, and remains the project's authoritative structured XAI study; whether the final Phase 2A model's maps would be rated differently has not been formally tested. All maps are **coarse contribution/sensitivity maps, not tumour segmentation or boundaries**.

11.1 APPROACH

Grad-CAM [7] explains only the differentiable neural head: maps were computed from the predicted-class score at `layer3[-1]` (14 × 14), `layer4[-1]` (7 × 7) and their 50 : 50 fusion. Because the deployed decision is made by the external SVM, **final-SVM occlusion** [8] was added to probe the complete deployed pipeline as a black box: a 32 × 32 patch (stride 16; 169 positions) set to the ImageNet mean was slid over the input, the complete CNN → LSTM → scaler → PCA → SVM pipeline was rerun, and the drop in the predicted class's decision value and any prediction change were recorded. The protocol was fixed before rating: seed 42, five test images per class (20; 15 tumour) for Grad-CAM, rated Good / Partial / Poor for localization; eleven of these (9 tumour, 2 no-tumour) chosen for occlusion before occlusion results were viewed (1,859 occlusions). Implementation details are in Appendix D.

11.2 FINDINGS

Table 11.1 Grad-CAM localization ratings for the 15 tumour images (Phase 1 ResNet152; per-case ratings in Appendix D).

Map	Good	Partial	Poor	Judged most relevant
Layer 3 (14 × 14)	1	9	5	9/15
Layer 4 (7 × 7)	3	2	10	6/15
Fused (50 : 50)	2	6	7	0/15

The neural head and SVM agreed on all 20 images and both were correct on 19; the shared error was Te-pi_0078 (pituitary predicted as glioma), with all maps rated Poor. Correct classification frequently coexisted with partial or poor localization: maps were often dominated by the skull margin, orbital or facial anatomy, extracranial regions or image borders. One notumor image (Te-no_0216) was flagged for expert review as a possible label anomaly; it was not relabelled.

Table 11.2 Final-SVM occlusion summary (Phase 1 ResNet152; per-case results in Appendix D).

Measure	Result
Tumour-case localization (Good / Partial / Poor)	2 / 2 / 5
More anatomically relevant: Grad-CAM / occlusion / both off-lesion	4 / 3 / 2
Class-changing occlusions (pooled)	58 / 1,859 (3.12%); median per image 0%; 5 of 11 images any flip
Misclassified Te-pi_0078 vs ten correct cases	38/169 (22.5%) vs 20/1,690 (1.18%)
Images whose maximum decision-value drop < 0.01	4 of 11

Interpretation and limits. High classification accuracy could not be equated with anatomically plausible or lesion-centred model reasoning. The study does not show which abstract features the network learned, and it does not prove shortcut learning. Occlusion probes the deployed SVM directly but was no more consistently lesion-focused than Grad-CAM; in no case were both clearly lesion-focused. Correct predictions were mostly stable under single-patch occlusion, while the misclassified case was fragile. Each map is normalized to its own maximum, so four images with negligible effects still look vivid: maps must be read with their effect sizes. Both studies are small (20 and 11 cases), single-rater and without radiologist review or expert masks. Two protocol images (Te-me_0012, a rated tumour image, and Te-no_0060, a no-tumour image) were later found among the material duplicates; removing them leaves the interpretation unchanged.

11.3 QUALITATIVE PROTOTYPE DEMONSTRATIONS

The final application can generate qualitative Grad-CAM and final-SVM occlusion for selected compatible frozen pipelines, including the ResNet152 Phase 2A representative (best_val_within_30, epoch 10). For ResNet models, layer 3, layer 4 and fused maps can be inspected; occlusion reruns the complete saved hybrid pipeline for 169 windows (32 × 32 patch, stride 16). These live views are not formally rated and do not constitute a Phase 2A or Phase 2B XAI experiment. The formally rated study remains Phase 1 ResNet152: it already answered RQ7 by showing that high accuracy does not necessarily imply lesion-centred reasoning. A second formal Phase 2 study was therefore not performed. Chapter 12 distinguishes these qualitative demonstrations from the saved Phase 1 findings; neither map type is tumour segmentation or localization ground truth.

CHAPTER 12
RESEARCH PROTOTYPE (PHASE 5)

The prototype initially demonstrated individual Phase 1 models. Once the study expanded to fixed 30-epoch training, sensitivity experiments, leakage auditing, frozen-pipeline external validation and explainability, that interface no longer represented the evidence needed to judge the models. The final Streamlit application [23] was therefore redesigned as an interactive research prototype and evidence-delivery interface over the completed study: it connects a selected prediction to its provenance, internal performance and limitations under external evaluation.

12.1 PURPOSE AND DESIGN OF THE FINAL PROTOTYPE

The application opens directly to Predict MRI (Fig. 12.1). Stage and backbone selection expose all 15 frozen pipelines: VGG16, VGG19, ResNet50, ResNet152 and EfficientNet-B3, each at Phase 1, Phase 2A and Phase 2B. These are independently selectable alternatives, not an ensemble; only the selected pipeline is used. The other 14 remain available as comparison evidence rather than being hidden behind the named representative.

The final representative remains ResNet152 — Phase 2A — best_val_within_30 — epoch 10: RMSprop, LR 1e-4, LSTM 256, dropout 0.3, Dense128, StandardScaler → PCA64 → linear SVM (C = 1). Its naming and selection caveat remain as described in Chapter 10. The interface exposes frozen artifacts and saved results without rerunning training.

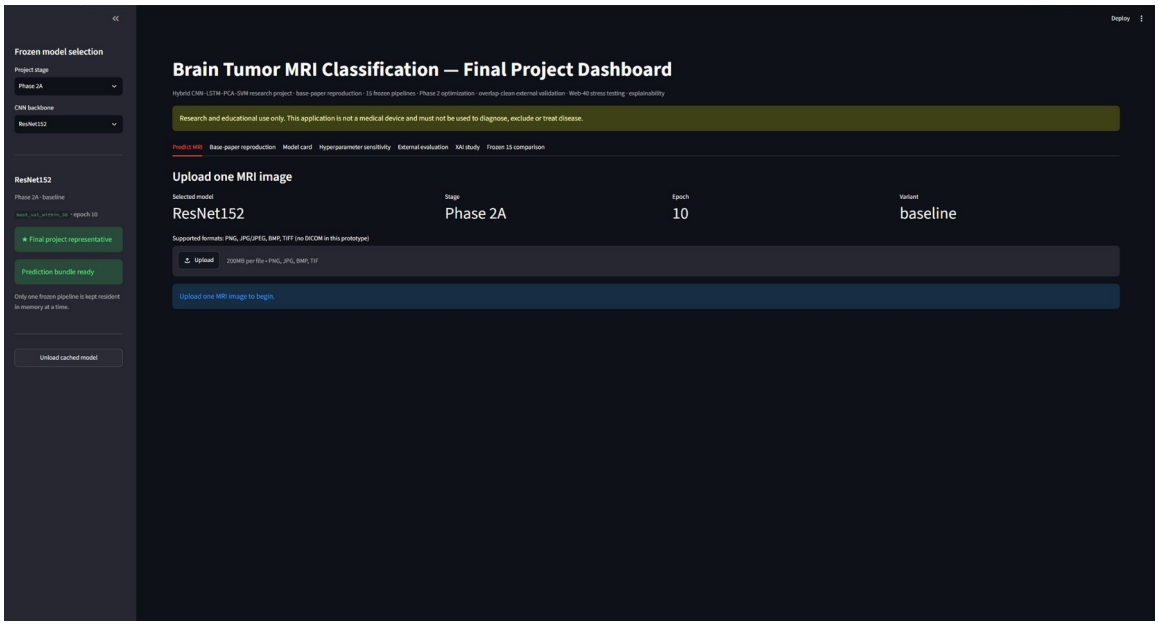


Fig. 12.1 Final Streamlit prediction interface.

Stage/backbone selection exposes 15 frozen Phase 1/2A/2B pipelines. Predict MRI is the landing tab; raster uploads use the saved hybrid bundle without refitting.

12.2 LIVE INFERENCE AND QUALITATIVE EXPLAINABILITY

One 2-D raster MRI image (PNG, JPG/JPEG, BMP or TIFF; no DICOM) passes through the deterministic evaluation transform: RGB conversion, direct resize to 224 × 224 and ImageNet normalization. No model is trained or altered, and no scaler, PCA or SVM is fitted on the upload. The final hybrid decision is the saved StandardScaler → PCA64 → linear SVM output after Dense128 feature extraction.

The prediction view reports final-SVM and neural-head classes, their agreement or disagreement, runtime, relative SVM class scores and neural softmax scores. The standardized SVMs use probability=False: decision-derived scores are relative scores, not calibrated probabilities. Neural softmax scores are also not clinically calibrated probabilities. Agreement permits inspection of the two classifier heads; it does not establish diagnostic reliability.

Live Grad-CAM explains the differentiable neural component of a compatible selected pipeline. ResNet views include layer 3, layer 4 and their fusion, raw heatmaps and MRI overlays with adjustable strength. Final-SVM occlusion complements this by rerunning MRI → CNN → spatial sequence → LSTM → Dense128 → StandardScaler → PCA64 → linear SVM. A 32 × 32 window at stride 16 gives 169 positions; the view reports the target class, window count, heatmap/overlay, prediction-flip rate, maximum SVM decision-score drop and runtime.

These live maps are qualitative contribution/sensitivity visualizations, not tumour segmentation, lesion masks, boundaries or radiological localization ground truth. Generating them for another stage does not extend the formally rated study beyond Phase 1 ResNet152 (Chapter 11).

12.3 INTERACTIVE ACCESS TO THE COMPLETED STUDY

Six further tabs (Figs. 12.2–12.7) make the saved research record inspectable. Base-paper reproduction places the reported architecture and RMSprop setup beside the implementation, split, Phase 1 neural-head/PCA+SVM comparison and Phase 2A follow-up. ResNet152 reached 98.40% internal test in Phase 1 and 98.703% in Phase 2A against the paper's approximately 99%; Phase 2A is the closest direct ResNet152 reproduction. EfficientNet-B3's 99.314% uses a different backbone. All these internal results retain the leakage caveat of Chapter 5.

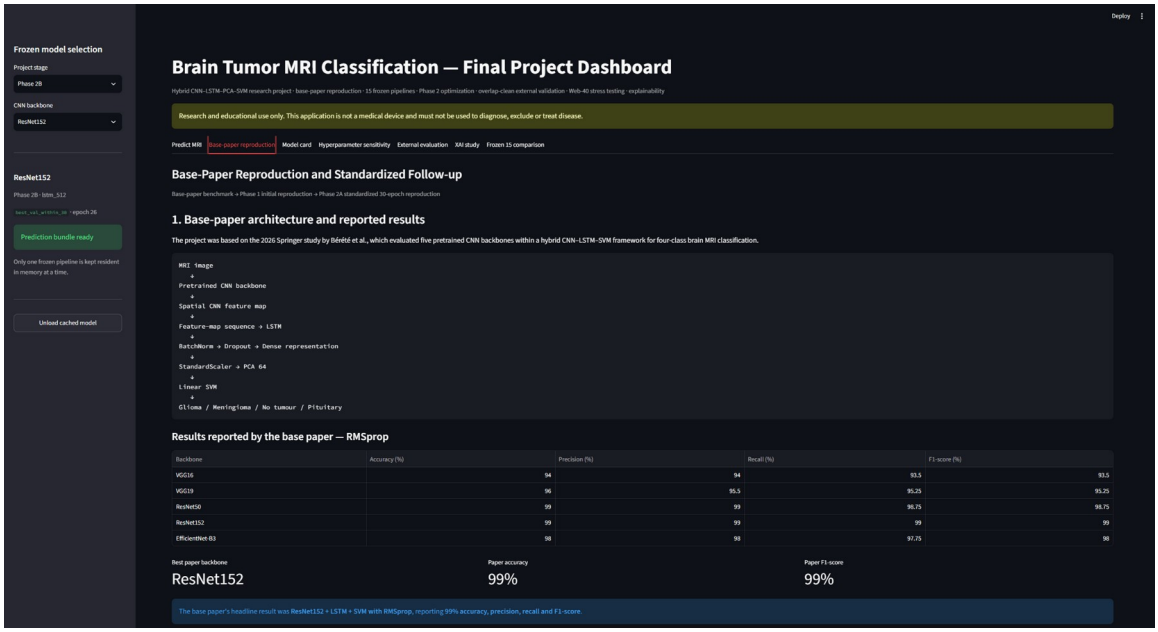


Fig. 12.2 Base-paper reproduction tab.

Architecture and reported RMSprop results appear within the saved research evidence interface; this view documents implementation, not a new experiment.

Model card links the selected pipeline to its variant, checkpoint/epoch, training configuration, classifier chain and artifact provenance, with available performance, confusion matrices, per-class results and training history.

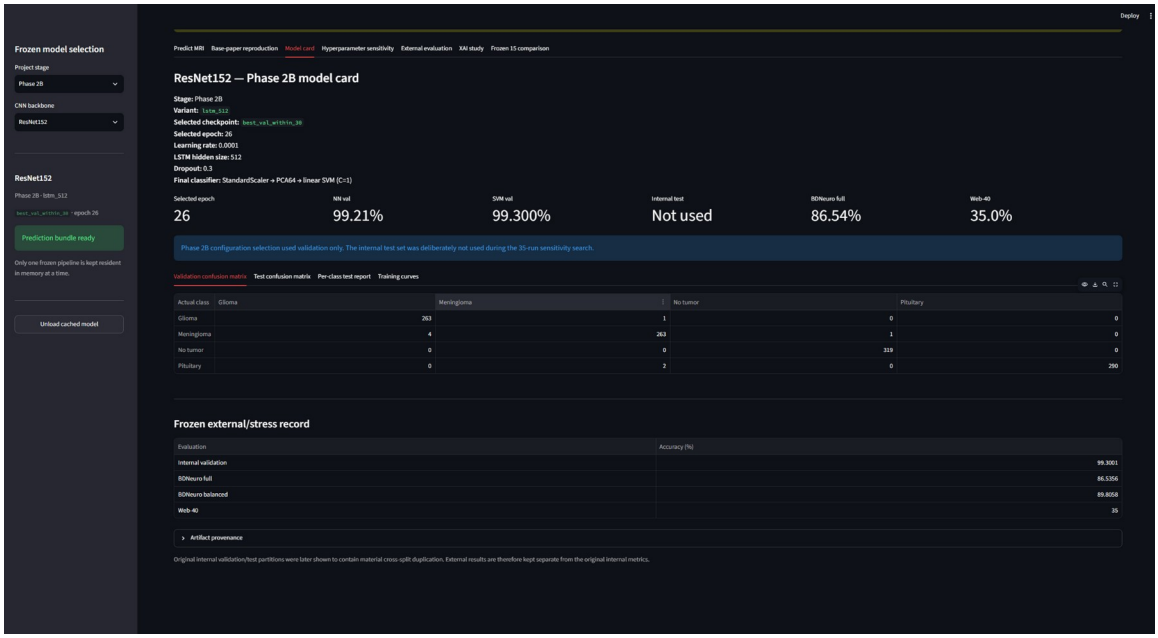


Fig. 12.3 Selected-pipeline model card.

Metadata, checkpoint results and external evidence remain visible. The final representative remains ResNet152 Phase 2A, epoch 10.

Hyperparameter sensitivity displays Phase 2B's one-factor changes: LR 1e-5, 5e-5 and 5e-4; LSTM 128 and 512; dropout 0.2 and 0.5. Five backbones × seven changes gave 35

runs and 70 downstream PCA+SVM evaluations across two retained states. The mean within-backbone ranges were 4.62 pp for LR, 0.58 pp for LSTM width and 0.42 pp for dropout; the best value depended on the backbone, and small internal gains did not consistently transfer.

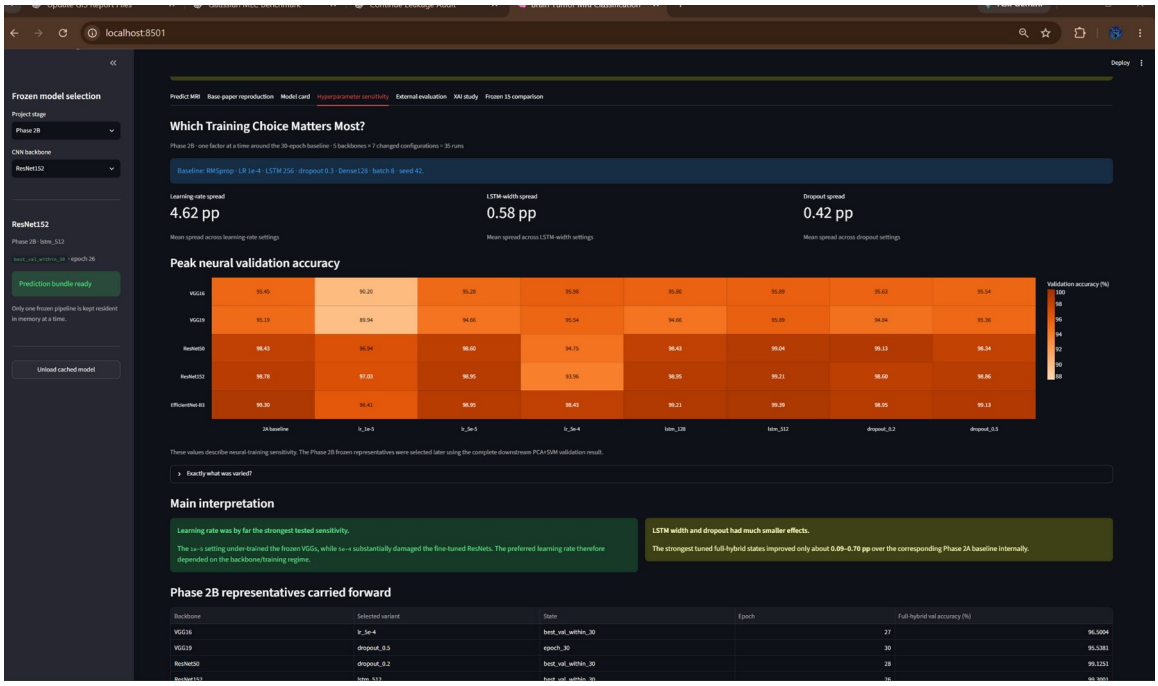


Fig. 12.4 Hyperparameter sensitivity tab.

Saved results from 35 one-factor runs show learning rate as the dominant tested sensitivity (4.62 pp versus 0.58/0.42 pp). No training is launched.

External evaluation (Fig. 12.5) begins with all 15 pipelines on overlap-clean BDNeuro full (n = 2,644), balanced BDNeuro (n = 1,648) and source-verified Web-40 (n = 40). It relates stage means and the representative's trajectory to external decline, no-tumour bias and the Nickparvar/BDNeuro integrity audits. Balanced BDNeuro is nested, not independent; Web-40 is exploratory. Neither source was used for retraining, and patient-level independence cannot be established.

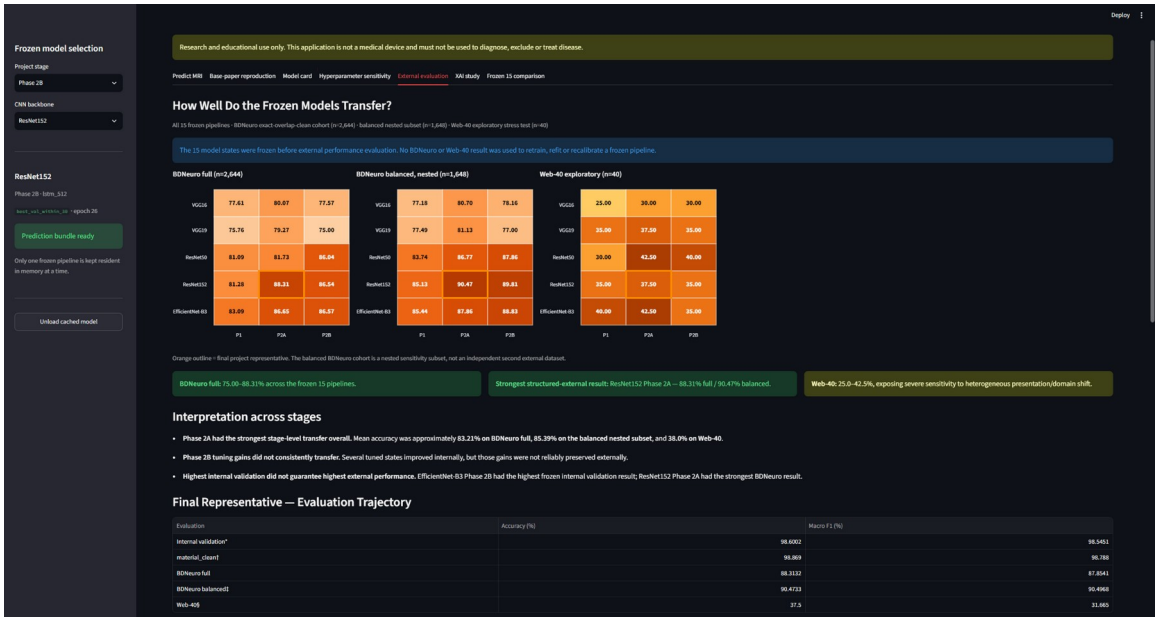


Fig. 12.5 External evaluation tab.

The 15 pipelines are compared on BDNeuro full, its nested balanced subset and exploratory Web-40, alongside interpretation and the representative trajectory.

XAI study displays the completed Phase 1 ResNet152 findings, not a new experiment: 20 Grad-CAM images (five/class; 15 tumour ratings) and 11 occlusion cases with 1,859 perturbations. Chapter 11 retains the ratings and case results, including 58 class changes and the fragile misclassified Te-pi_0078. High accuracy did not imply consistently lesion-centred reasoning; shortcut learning was not proven. The saved protocol and results are shown in Fig. 12.6.

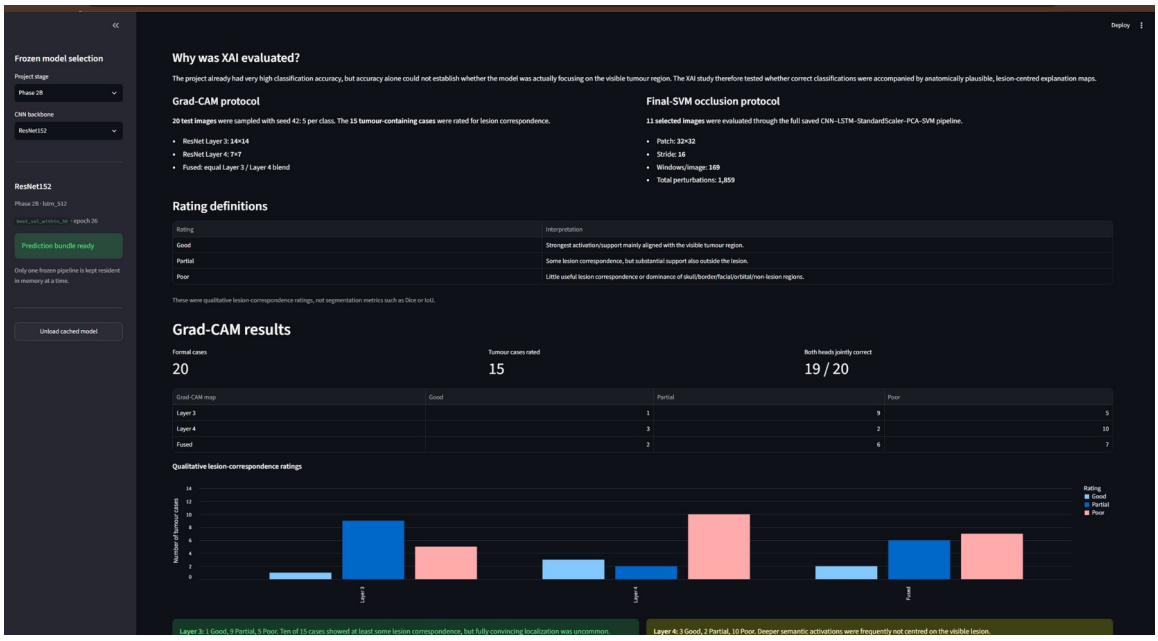


Fig. 12.6 Formal XAI study interface.

Saved findings are displayed; no second formal XAI study is conducted.

Frozen 15 comparison permits stage/backbone filtering of internal validation, BDNeuro full/balanced and Web-40 metrics, keeping the highest internal score distinct from external robustness.

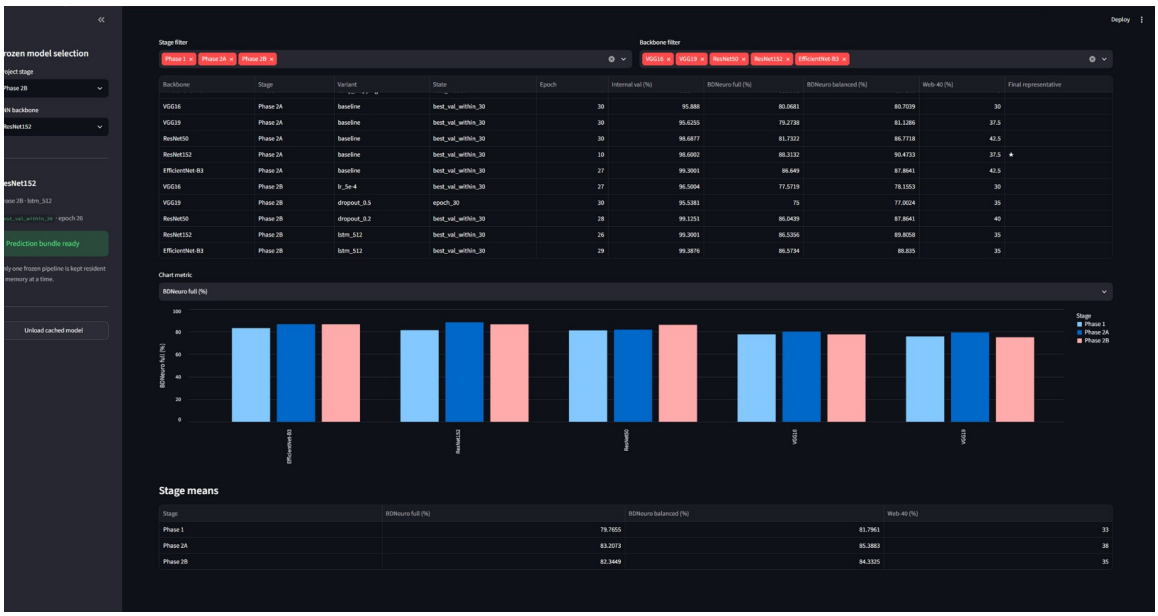


Fig. 12.7 Frozen 15 comparison tab.

Backbone/stage filters, the complete comparison table and stage summaries expose independently selectable frozen pipelines, not an ensemble.

12.4 INTERPRETATION AND SCOPE

Interpretation and consequence. The final prototype exposes the project's evidence trajectory — reproduction, optimization, external failure and explainability — through one reproducible interface. It does not make the strongest internal model a diagnostic system: a model should not be judged from one headline accuracy. This remains a research/educational prototype, not a medical device; it has no clinical validation and must not be used for diagnosis. Saved results answer the research questions; uploads and qualitative maps demonstrate the frozen workflow.

CHAPTER 13
DISCUSSION

13.1 ANSWERS TO THE RESEARCH QUESTIONS

Table 13.1 The research questions and the answers the evidence supports.

Question	Answer	Evidence
RQ1: Can the base-paper hybrid architecture achieve strong four-class performance?	Yes, within the dataset, across five pipelines (93.52–98.47% internal* test); the 99% headline was not reproduced exactly.	Ch. 4
RQ1: Does PCA + SVM add value over the neural head?	On test, for every backbone (+0.23 to +1.30 pp); not consistently on validation. No universal benefit claimed.	Ch. 4
RQ2: Can the ~98–99% public-split result be read as independent generalization?	Not safely: 562 material cross-split pairs (250 test images); patient-level independence cannot be established.	Ch. 5
RQ2: Are the high scores explained only by duplicated test images?	No: material_clean stayed within 0.5 pp; training/validation contamination remains unresolved.	Ch. 5
RQ3: Was early stopping hiding convergence?	For the VGGs, yes: they kept improving to epoch 30; every backbone's selected full-hybrid validation exceeded its Phase 1 result.	Ch. 6
RQ4: Which training choice matters most, and does more optimization guarantee better generalization?	Learning rate (4.62 pp mean range). No: small Phase 2B internal gains did not consistently transfer externally.	Ch. 7, 10
RQ5: Do frozen models generalize to a different structured dataset?	Partially: 75.00–88.31% on exact-overlap-clean BDNeuro, 10.29–20.54 pp below internal validation.	Ch. 8
RQ6: Are they robust to severe real-world presentation shift?	Not in the exploratory Web-40 test: 25.0–42.5%, with a strong no-tumour prediction bias.	Ch. 9
RQ7: Does correct classification imply lesion-centred reasoning?	No: the formal XAI study did not provide consistent evidence that correct predictions were lesion-centred; they often coexisted with partial or poor localization.	Ch. 11

* known duplicate-contaminated original split.

13.2 AN INTEGRATED READING

Each experiment answered a weakness left by the previous one, and together they form one argument. Phase 1 showed that the pipeline works; the audit showed that the evidence for that claim rested on a contaminated split; material_clean showed that the internal result was not merely an artefact of duplicated test images; Phases 2A and 2B characterized training behaviour and sensitivity; the freeze made external evaluation tamper-evident; BDNeuro measured structured transfer after objective overlap removal; Web-40 probed robustness to severe shift; and explainability tested whether correct predictions rest on plausible image evidence. **Every layer that moved further from the training collection lowered confidence in the headline number**, while none of them showed the models to be useless.

BDNeuro represents moderate shift (a different source but a curated dataset of similar format); the web set represents severe shift. The step-wise drop from about 93–99% internally to 75–88% to 25–42.5% is consistent with models that learned the appearance of their training collection — intensity profile, crop style, slice selection, sequence mix — as well as tumour morphology. The no-tumour bias on web images and the frequent off-lesion attention point the same way, but without acquisition metadata these remain plausible hypotheses, not demonstrated causes. The tuning result adds a methodological point: internal gains of fractions of a point on one contaminated validation split were not a reliable guide to robustness.

Leakage appeared twice. About 19% of the official test images and nearly 8% of training images take part in a material cross-split pair, so any accuracy on this split is partly a within-collection estimate that may be optimistic. A second, larger overlap appeared externally: 3,297 BDNeuro images were exact pixel copies of development images. The substantial exact overlap observed between these two public collections shows that public medical-image datasets can contain redistributed or repackaged images, so image-overlap auditing is necessary before treating a public dataset as independent external evidence; external evaluation should also be frozen in advance so that it cannot be tuned against.

13.3 COMPARISON WITH THE BASE PAPER

Table 13.2 This project versus the base paper (RMSprop).

Backbone	Paper [1]	P1 test*	P1 material_clean	P2A test*	Best BDNeuro full
VGG16	94%	94.97%	94.816%	96.186%	80.07%
VGG19	96%	93.52%	93.025%	95.271%	79.27%
ResNet50	99%	98.09%	98.303%	98.932%	86.04%
ResNet152	99%	98.40%	98.398%	98.703%	88.31%
EfficientNet-B3	98%	98.47%	98.680%	99.314%	86.65%

Project internal numbers are on the same public split. * Original internal test = known duplicate-contaminated split (Chapter 5); never used for selection. material_clean = post-hoc leakage-aware subset.

Across Phase 1 and Phase 2A, the project's internal test accuracy differs from the paper's figures by 0.07–0.91 pp for the ResNets (always below 99%), 0.47–1.31 pp for EfficientNet-B3 (above), 0.97–2.19 pp for VGG16 (above) and 0.73–2.48 pp for VGG19 (below). The 99% headline is **not reproduced** at the same number; plausible contributors are batch size 8 versus 32, early stopping in Phase 1, the frozen-VGG policy and RMSprop-only training. More importantly, the project measures what the paper does not: the shared public split contains material duplicate leakage, structured external accuracy for the same pipelines is 10.29–20.54 pp lower, and the formal XAI study did not provide consistent evidence that correct predictions were lesion-centred. The base paper's result remains a within-dataset benchmark; this project shows how much that benchmark leaves unanswered.

CHAPTER 14
LIMITATIONS

1. **Contaminated development split.** 562 material cross-split pairs; all internal results (Phase 1, 2A, 2B) are historical results on this split.
2. **material_clean is post-hoc.** It removes implicated test images only, does not repair training or validation and changes class composition. Duplicate-group-aware re-splitting and retraining were not performed.
3. **No patient-level independence** for Nickparvar, BDNeuro or the web images (no reliable patient identifiers).
4. **Single seed and split.** No repeated runs, confidence intervals or significance tests; small differences, including all Phase 2B gains, are not statistically established.
5. **Unequal training policy.** Frozen VGG versus fine-tuned others and unequal Phase 1 budgets make cross-backbone comparisons pipeline comparisons.
6. **Limited coverage and ablation.** Three factors varied one at a time with RMSprop only; no no-LSTM, no-PCA or alternative-ordering ablation.
7. **Simple preprocessing.** Direct resize without skull stripping or cropping lets the model use skull, background and border cues.
8. **External cohort limits.** Exact-pixel cleaning improves image-level but not patient-level independence; non-identical pHash candidates were retained by design; the BDNeuro balanced subset is nested; the BDNeuro licence/citation fields are unfilled.
9. **Selection optimism for the named representative**, whose BDNeuro figures informed its naming (no model was changed).
10. **Web-40 scale.** n = 40, one image = 2.5 pp, publication images, source-page diagnoses; exploratory only.
11. **Explainability and prototype scope.** The formal XAI studies are small, single-rater and on the Phase 1 ResNet152 only (live GUI maps for selected frozen models are qualitative); the prototype accepts one 2-D raster image only (no DICOM), performs no refitting on uploads and shows uncalibrated SVM/softmax scores; XAI is not segmentation. It is a research/educational prototype, not a medical device or a system for diagnosis. Historical figures without machine-readable records (VGG16 collapse ≈ 28%, legacy prototype ≈ 97.64%, NOR_01 pre-replacement audit; Appendix F) are labelled as such. No clinical validation of any kind.

CHAPTER 15
CONCLUSION AND FUTURE WORK

This project re-implemented the hybrid CNN–LSTM–PCA–SVM architecture of Bérété et al. (2026) with five independent backbones and asked how far its near-99% within-dataset accuracy can be trusted. Internally the pipelines reached 93.5–98.5% test accuracy, but a duplicate audit found material cross-split leakage (562 pairs; 250 test images), so these are historical results on a contaminated split; a post-hoc material_clean evaluation showed they were not driven solely by duplicated test images. Under a fixed 30-epoch protocol every backbone's selected full-hybrid validation exceeded its Phase 1 result, and learning rate was the dominant tested sensitivity (4.62 pp mean range). The 15 pipelines were selected using internal validation only, frozen and SHA-256 hashed before external performance evaluation; after removing 3,297 pixel-identical BDNeuro images, accuracy was 75.0–88.3% on the 2,644-image cohort and 25–42.5% on the locked, exploratory Web-40 set, with most tumours called normal, and small internal tuning gains did not consistently transfer. The final representative, ResNet152 Phase 2A (best_val_within_30, epoch 10), traces the whole trajectory: 98.60% internal validation, 98.70% internal test, 98.869% material_clean, 88.31% BDNeuro full, 90.47% BDNeuro balanced (figures that informed its naming, so they carry some selection optimism) and 37.5% on Web-40.

The central conclusion is that high performance on the original public dataset did not guarantee robust out-of-distribution performance. A second conclusion follows from the explainability study: **high classification accuracy could not be equated with anatomically plausible or lesion-centred model reasoning.** The project's contribution is layered, auditable evidence of how and where performance degrades. The system remains a research and educational prototype.

Future work (post-final, not completed work). Any change motivated by the BDNeuro or web results must be evaluated on a *new*, untouched evaluation set, because those sets have now been seen.

- **Cleaner and stronger evidence:** group duplicates, re-split by group and retrain all five backbones; evaluate on patient-identified, multi-site and DICOM/volumetric cohorts; repeat with several seeds and report confidence intervals.
- **Understanding the architecture:** no-LSTM, no-PCA and alternative-ordering ablations; calibration.
- **Robustness, explainability and software:** MRI-specific normalization, skull stripping and domain adaptation on new external data; XAI scored against expert lesion masks; DICOM ingestion; the base paper's remaining optimizers.

REFERENCES

[1] M. Bérété, A. Echtioui, L. Sellami and A. A. Ben Hmida, "Hybrid CNN+LSTM+SVM model for advanced brain tumor classification," *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, art. 43, 2026. doi:10.1007/s13721-025-00718-8.

[2] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015. arXiv:1409.1556.

[3] K. He, X. Zhang, S. Ren and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778. doi:10.1109/CVPR.2016.90.

[4] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, PMLR 97, 2019, pp. 6105–6114.

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997. doi:10.1162/neco.1997.9.8.1735.

[6] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, pp. 273–297, 1995. doi:10.1007/BF00994018.

[7] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, no. 2, pp. 336–359, 2020. doi:10.1007/s11263-019-01228-7.

[8] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. ECCV*, 2014, pp. 818–833. doi:10.1007/978-3-319-10590-1_53.

[9] M. S. Başarslan, "M-C&M-BL: a novel classification model for brain tumor classification: multi-CNN and multi-BiLSTM," *Journal of Supercomputing*, vol. 81, art. 502, 2025. doi:10.1007/s11227-025-06964-x.

[10] F. A. Özbay and E. Özbay, "Brain tumor detection with mRMR-based multimodal fusion of deep learning from MR images using Grad-CAM," *Iran Journal of Computer Science*, vol. 6, pp. 245–259, 2023. doi:10.1007/s42044-023-00137-w.

[11] K. M. Hosny, M. A. Mohammed, R. A. Salama and A. M. Elshewey, "Explainable ensemble deep learning-based model for brain tumor detection and classification," *Neural Computing and Applications*, vol. 37, pp. 1289–1306, 2025. doi:10.1007/s00521-024-10401-0.

[12] M. M. Musthafa, T. R. Mahesh, V. Vinoth Kumar and S. Guluwadi, "Enhancing brain tumor detection in MRI images through explainable AI using Grad-CAM with Resnet 50," *BMC Medical Imaging*, vol. 24, art. 107, 2024. doi:10.1186/s12880-024-01292-7.

[13] S. Panigrahi, D. R. D. Adhikary and B. K. Pattanayak, "Brain tumor classification: a blend of ensemble learning and fine-tuned pre-trained models," *Discover Applied Sciences*, 2025. doi:10.1007/s42452-025-06695-x.

[14] S. Sahoo, S. Mishra, B. Brahma, P. Barsocchi and A. K. Bhoi, "SSO-CCNN: a correlation-based optimized deep CNN for brain tumor classification using sampled PGGAN," *International Journal of Computational Intelligence Systems*, vol. 17, 2024. doi:10.1007/s44196-024-00574-w.

[15] R. L. Kumar, J. Kakarla, B. V. Isunuri and M. Singh, "Multi-class brain tumor classification using residual network and global average pooling," *Multimedia Tools and Applications*, vol. 80, no. 9, pp. 13429–13438, 2021. doi:10.1007/s11042-020-10335-4.

[16] M. Nickparvar, "Brain Tumor MRI Dataset," Kaggle. https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset.

[17] "Brain Tumor MRI Dataset (Glioma, Meningioma, Pituitary, No Tumor)" — BDNeuro-MRI, Version 8 (project copy README; source stated as "Epic & CSCR Hospital"). The licence and citation fields of the distributed README are unfilled.

[18] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE CVPR*, 2009, pp. 248–255.

[19] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. ICML*, 2015, pp. 448–456.

[20] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *JMLR*, vol. 15, pp. 1929–1958, 2014.

[21] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[22] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[23] Streamlit open-source app framework. https://streamlit.io

APPENDIX A

THE 15 FROZEN PIPELINES: COMPLETE EXTERNAL RESULTS

All 15 pipelines were selected using internal validation only, frozen and SHA-256 hashed before external performance evaluation. Gap = external accuracy − internal full-hybrid validation accuracy. State: BM = best_model (Phase 1), B30 = best_val_within_30, E30 = epoch_30.

Fig. A.1 plots each frozen pipeline's internal full-hybrid validation against its BDNeuro full-cohort accuracy.

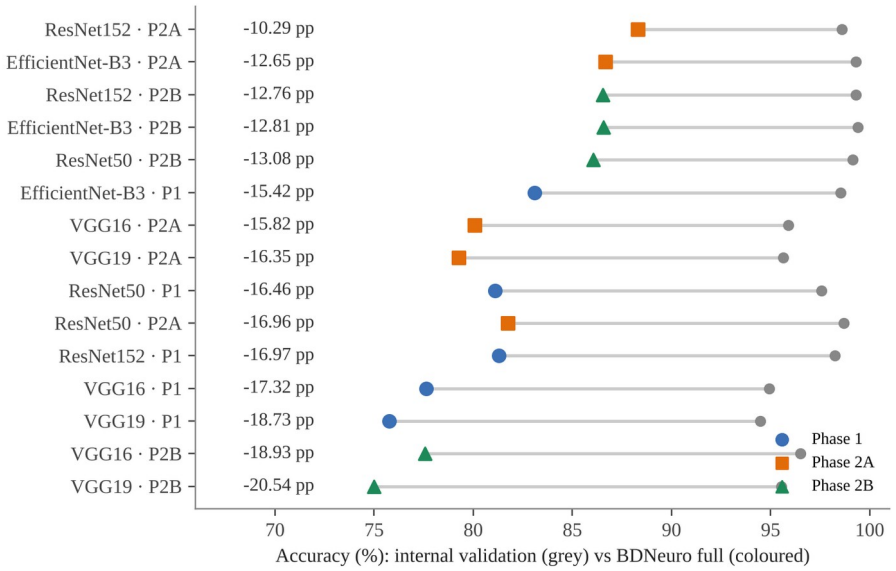


Fig. A.1 Internal full-hybrid validation (grey) versus BDNeuro full-cohort accuracy for each frozen pipeline, sorted by gap (values in Table A.1).

Table A.1 Complete external results of the 15 frozen pipelines (%; gaps in pp).

Backbone	Stage	Variant	State · ep	Int. val*	Full acc	Full F1	Full gap	Bal. acc	Bal. F1	Bal. gap	Web acc (n)	Web F1	Web gap
VGG16	P1	early_stopping	BM · 16	94.926%	77.61	75.50	−17.32	77.18	76.82	−17.74	25.0 (10/40)	14.58	−69.93
VGG16	P2A	baseline	B30 · 30	95.888%	80.07	78.71	−15.82	80.70	80.62	−15.18	30.0 (12/40)	23.23	−65.89
VGG16	P2B	lr_5e-4	B30 · 27	96.500%	77.57	76.01	−18.93	78.16	77.88	−18.35	30.0 (12/40)	21.59	−66.50
VGG19	P1	early_stopping	BM · 18	94.488%	75.76	74.80	−18.73	77.49	77.64	−17.00	35.0 (14/40)	28.89	−59.49
VGG19	P2A	baseline	B30 · 30	95.626%	79.27	78.12	−16.35	81.13	81.24	−14.50	37.5 (15/40)	27.78	−58.13
VGG19	P2B	dropout_0.5	E30 · 30	95.538%	75.00	74.27	−20.54	77.00	77.15	−18.54	35.0 (14/40)	25.45	−60.54
ResNet50	P1	early_stopping	BM · 7	97.550%	81.09	79.93	−16.46	83.74	83.80	−13.81	30.0 (12/40)	26.49	−67.55
ResNet50	P2A	baseline	B30 · 30	98.688%	81.73	81.42	−16.96	86.77	86.90	−11.92	42.5 (17/40)	38.18	−56.19
ResNet50	P2B	dropout_0.2	B30 · 28	99.125%	86.04	84.83	−13.08	87.86	87.87	−11.26	40.0 (16/40)	33.91	−59.13
ResNet152	P1	early_stopping	BM · 8	98.250%	81.28	80.84	−16.97	85.13	85.13	−13.12	35.0 (14/40)	28.08	−63.25
ResNet152	P2A	baseline	B30 · 10	98.600%	88.31	87.85	−10.29	90.47	90.50	−8.13	37.5 (15/40)	31.66	−61.10
ResNet152	P2B	lstm_512	B30 · 26	99.300%	86.54	85.67	−12.76	89.81	89.76	−9.49	35.0 (14/40)	25.74	−64.30
EfficientNet-B3	P1	early_stopping	BM · 6	98.513%	83.09	81.93	−15.42	85.44	85.60	−13.08	40.0 (16/40)	34.78	−58.51
EfficientNet-B3	P2A	baseline	B30 · 27	99.300%	86.65	85.12	−12.65	87.86	87.79	−11.44	42.5 (17/40)	31.97	−56.80
EfficientNet-B3	P2B	lstm_512	B30 · 29	99.388%	86.57	85.53	−12.81	88.83	88.96	−10.55	35.0 (14/40)	28.07	−64.39

BDNeuro balanced = nested sensitivity subset; Web-40 = exploratory stress test. * internal full-hybrid validation on the known duplicate-contaminated original split. Baseline = lr1e-4_lstm256_dropout0.3.

APPENDIX B
PHASE 2A AND PHASE 2B DETAIL

Phase 2A trained each baseline configuration for exactly 30 epochs; Phase 2B trained seven one-factor configurations per backbone (35 runs, each for 30 epochs) and evaluated both retained states downstream (70 PCA + SVM evaluations). Phase 2B never used the test set.

Table B.1 Phase 2A full detail.

Backbone	Best neural epoch	Neural best val	Neural val ep 30	SVM val	SVM val F1	SVM val (epoch_30)	Test*	material_ clean*
VGG16	30	95.45%	95.45%	95.888%	95.746%	95.888%	96.186%	95.853%
VGG19	30	95.19%	95.19%	95.626%	95.462%	95.626%	95.271%	95.099%
ResNet50	30	98.43%	98.43%	98.688%	98.644%	98.688%	98.932%	98.963%
ResNet152	10	98.78%	98.69%	98.600%	98.545%	98.513%	98.703%	98.869%
EfficientNet-B3	27	99.30%	98.69%	99.300%	99.270%	99.213%	99.314%	99.246%

Neural = neural-head validation accuracy; SVM = full hybrid pipeline. Representative state = best_val_within_30 for all five backbones. * Original internal test = known duplicate-contaminated split (Chapter 5); never used for selection. material_clean = post-hoc leakage-aware subset (n = 1,061).

Table B.2 Phase 2A states on the material_clean post-hoc leakage-aware subset (n = 1,061; 250 excluded).

Backbone	State	Epoch	Full test*	material_clean acc	material_clean F1	Δ (pp)
VGG16	best_val_within_30	30	96.186%	95.853%	95.442%	−0.333
VGG19	best_val_within_30	30	95.271%	95.099%	94.655%	−0.172
ResNet50	best_val_within_30	30	98.932%	98.963%	98.848%	+0.031
ResNet152	best_val_within_30	10	98.703%	98.869%	98.788%	+0.166
ResNet152	epoch_30	30	98.627%	98.680%	98.598%	+0.053
EfficientNet-B3	best_val_within_30	27	99.314%	99.246%	99.148%	−0.068
EfficientNet-B3	epoch_30	30	99.161%	99.246%	99.151%	+0.085

Full = original 1,311-image test *of the same state.* known duplicate-contaminated original split.

Table B.3 Selected Phase 2B representatives (highest full-hybrid validation among 14 states per backbone).

Backbone	Configuration	State	Epoch	SVM val	SVM val F1	Phase 2A SVM val
VGG16	lr_5e-4	best_val_within_30	27	96.500%	96.321%	95.888%
VGG19	dropout_0.5	epoch_30	30	95.538%	95.382%	95.626%
ResNet50	dropout_0.2	best_val_within_30	28	99.125%	99.122%	98.688%
ResNet152	lstm_512	best_val_within_30	26	99.300%	99.265%	98.600%
EfficientNet-B3	lstm_512	best_val_within_30	29	99.388%	99.377%	99.300%

Table B.4 All 35 Phase 2B runs: neural (NN) peak and epoch-30 validation, late-epoch drift, and full-hybrid SVM validation of both retained states (%).

Backbone	Run	Best ep.	NN peak val	NN val ep 30	Drift (pp)	SVM val (best state)	SVM val (epoch_30)
VGG16	lr_1e-5	30	90.20	90.20	+0.00	91.514	91.514
VGG16	lr_5e-5	30	95.28	95.28	+0.00	95.188	95.188
VGG16	lr_5e-4	27	95.98	95.98	+0.00	96.500	95.976
VGG16	lstm_128	30	95.80	95.80	+0.00	95.626	95.626
VGG16	lstm_512	19	95.89	95.63	-0.26	95.363	95.538
VGG16	dropout_0.2	27	95.63	95.54	-0.09	95.538	96.413
VGG16	dropout_0.5	30	95.54	95.54	+0.00	95.801	95.801
VGG19	lr_1e-5	30	89.94	89.94	+0.00	89.764	89.764
VGG19	lr_5e-5	28	94.66	94.05	-0.61	94.226	94.401
VGG19	lr_5e-4	27	95.54	95.01	-0.52	95.451	95.188
VGG19	lstm_128	28	94.66	94.31	-0.35	95.276	94.401
VGG19	lstm_512	30	95.89	95.89	+0.00	95.276	95.276
VGG19	dropout_0.2	29	94.84	94.75	-0.09	94.751	95.101
VGG19	dropout_0.5	28	95.36	95.19	-0.17	95.451	95.538
ResNet50	lr_1e-5	27	96.94	96.76	-0.17	97.638	96.850
ResNet50	lr_5e-5	23	98.60	98.43	-0.17	98.600	98.688
ResNet50	lr_5e-4	25	94.75	90.99	-3.76	95.888	94.838
ResNet50	lstm_128	28	98.43	96.68	-1.75	98.425	97.900
ResNet50	lstm_512	29	99.04	98.25	-0.79	98.863	98.688
ResNet50	dropout_0.2	28	99.13	98.51	-0.61	99.125	98.513
ResNet50	dropout_0.5	26	98.34	97.46	-0.87	98.688	97.988
ResNet152	lr_1e-5	29	97.03	96.94	-0.09	97.113	97.375
ResNet152	lr_5e-5	17	98.95	98.51	-0.44	99.213	98.600
ResNet152	lr_5e-4	20	93.96	93.61	-0.35	95.101	94.488
ResNet152	lstm_128	29	98.95	98.69	-0.26	98.950	98.688
ResNet152	lstm_512	26	99.21	98.78	-0.44	99.300	98.688
ResNet152	dropout_0.2	27	98.60	98.08	-0.52	99.125	98.425
ResNet152	dropout_0.5	29	98.86	98.43	-0.44	98.775	98.513
EfficientNet-B3	lr_1e-5	28	96.41	96.33	-0.09	96.938	97.375
EfficientNet-B3	lr_5e-5	14	98.95	98.95	+0.00	98.950	98.600
EfficientNet-B3	lr_5e-4	27	98.43	97.64	-0.79	98.600	98.425
EfficientNet-B3	lstm_128	30	99.21	99.21	+0.00	99.038	99.038
EfficientNet-B3	lstm_512	29	99.39	98.43	-0.96	99.388	98.863
EfficientNet-B3	dropout_0.2	15	98.95	98.69	-0.26	98.950	98.863
EfficientNet-B3	dropout_0.5	26	99.13	98.69	-0.44	99.125	98.600

No test metrics exist by design.

APPENDIX C

PER-CLASS RECALL OF THE 15 FROZEN PIPELINES

Table C.1 Per-class recall (%) from each pipeline's confusion matrix: BDNeuro full cohort (n = 2,644) and Web-40 exploratory stress test (10 images per class; one image = 10 pp of class recall).

Backbone	Stage	BD glioma	BD mening.	BD notumor	BD pituit.	Web glioma	Web mening.	Web notumor	Web pituit.
VGG16	P1	82.7	75.0	94.7	58.1	10	0	90	0
VGG16	P2A	82.5	85.8	93.7	61.6	20	0	90	10
VGG16	P2B	79.0	81.8	94.7	59.7	10	20	90	0
VGG19	P1	72.3	82.4	90.0	67.4	10	30	90	10
VGG19	P2A	77.4	85.0	90.0	71.0	10	40	100	0
VGG19	P2B	72.2	82.2	89.8	64.6	10	30	100	0
ResNet50	P1	76.9	90.0	98.5	70.2	10	20	70	20
ResNet50	P2A	73.9	92.2	98.8	77.0	20	80	60	10
ResNet50	P2B	83.4	92.8	98.1	77.5	20	30	100	10
ResNet152	P1	72.8	93.2	97.8	76.9	20	10	100	10
ResNet152	P2A	83.5	94.9	93.2	89.0	10	40	90	10
ResNet152	P2B	82.0	96.2	97.6	80.2	0	40	90	10
EfficientNet-B3	P1	79.7	89.6	98.1	74.2	10	30	100	20
EfficientNet-B3	P2A	86.6	94.3	96.8	73.7	10	60	100	0
EfficientNet-B3	P2B	83.3	95.1	99.0	77.5	20	30	90	0

APPENDIX D
EXPLAINABILITY CASE DETAIL

All cases below belong to the formal Phase 1 ResNet152 study (20 Grad-CAM images, 15 of them tumour images rated; 11 occlusion images × 169 windows = 1,859 occlusion runs, 58 class flips; Te-pi_0078: 38/169 flips). They do not describe the final Phase 2A representative, whose application maps are qualitative only. All maps are coarse contribution/sensitivity maps, not tumour segmentation.

Implementation. Grad-CAM weights each activation map of the chosen layer by the mean gradient of the neural head's predicted-class score, keeps the positive part and upsamples to 224 × 224. To let gradients flow through the LSTM, the LSTM alone is switched to training mode with cuDNN disabled, while BatchNorm and Dropout stay in evaluation mode. Occlusion replaces each 32 × 32 window with the ImageNet channel mean in normalized space and reruns the complete saved pipeline.

Table D.1 Grad-CAM ratings for the 20-image protocol (Phase 1 ResNet152, seed 42).

Image	True	SVM pred.	Layer 3	Layer 4	Fused	Most relevant
Te-glTr_0002	glioma	glioma	Partial	Poor	Poor	Layer 3
Te-gl_0030	glioma	glioma	Poor	Poor	Poor	Layer 4
Te-gl_0085	glioma	glioma	Poor	Poor	Poor	Layer 3
Te-gl_0148	glioma	glioma	Partial	Poor	Poor	Layer 3
Te-gl_0282	glioma	glioma	Poor	Partial	Partial	Layer 4
Te-me_0012	meningioma	meningioma	Good	Good	Good	Layer 4
Te-me_0118	meningioma	meningioma	Partial	Poor	Poor	Layer 3
Te-me_0137	meningioma	meningioma	Partial	Good	Good	Layer 4
Te-me_0155	meningioma	meningioma	Partial	Partial	Partial	Layer 4
Te-me_0204	meningioma	meningioma	Partial	Good	Partial	Layer 4
Te-no_0060	notumor	notumor	n/a	n/a	n/a	n/a
Te-no_0157	notumor	notumor	n/a	n/a	n/a	n/a
Te-no_0191	notumor	notumor	n/a	n/a	n/a	n/a
Te-no_0216	notumor	notumor	n/a	n/a	n/a	n/a
Te-no_0352	notumor	notumor	n/a	n/a	n/a	n/a
Te-piTr_0008	pituitary	pituitary	Partial	Poor	Partial	Layer 3
Te-pi_0010	pituitary	pituitary	Poor	Poor	Poor	Layer 3
Te-pi_0078	pituitary	glioma	Poor	Poor	Poor	Layer 3
Te-pi_0267	pituitary	pituitary	Partial	Poor	Partial	Layer 3
Te-pi_0282	pituitary	pituitary	Partial	Poor	Partial	Layer 3

Coarse contribution maps, not segmentation; n/a = no-tumour image.

Table D.2 Final-SVM occlusion per case (169 occlusions per image).

Image	True	Occlusion rating	Comparison with Grad-CAM	Flips	Max score drop
Te-glTr_0002	glioma	Poor	Both off-lesion	0/169	0.0051
Te-gl_0030	glioma	Partial	Occlusion more relevant	6/169	1.4187
Te-gl_0282	glioma	Good	Occlusion more relevant	5/169	1.5125
Te-me_0012	meningioma	Poor	Grad-CAM more relevant	0/169	0.0016
Te-me_0118	meningioma	Poor	Grad-CAM more relevant	8/169	1.0422
Te-me_0137	meningioma	Partial	Grad-CAM more relevant	0/169	0.0045
Te-no_0060	notumor	n/a	n/a (no tumour)	0/169	0.0190
Te-no_0352	notumor	n/a	n/a (no tumour)	0/169	0.0061
Te-piTr_0008	pituitary	Poor	Grad-CAM more relevant	0/169	0.0149
Te-pi_0078	pituitary	Poor	Both off-lesion	38/169	2.5324
Te-pi_0267	pituitary	Good	Occlusion more relevant	1/169	1.0409

Flips = occlusions that changed the predicted class.

APPENDIX E
EVALUATION DESIGN, TECHNICAL IMPLEMENTATION AND EVIDENCE SOURCES

Table E.1 Evaluation design by research question.

RQ	Evaluation	Data	Safeguard
1	Neural head vs PCA + SVM on the same checkpoints	Nickparvar validation and test*	Selection on validation only; test never used for selection
2	Cross-split duplicate audit; post-hoc test policies	All 7,023 Nickparvar images	Policies defined before re-scoring; nothing retrained
3, 4	Fixed 30-epoch horizon; one-factor sensitivity	Nickparvar training/validation	Phase 2B never evaluated on the test set
5	Frozen 15-pipeline external evaluation	BDNeuro after its own overlap audit	Pipelines hashed before external evaluation; no refitting
6	In-the-wild stress test	Web-40, source-verified	Audited and locked before inference
7	Grad-CAM [7] and final-SVM occlusion [8], rated	Phase 1 ResNet152, 20 / 11 test images	Case selection and rating protocol fixed before results

* known duplicate-contaminated original split.

Table E.2 Class distribution of the Nickparvar development dataset.

Class	Training folder	Testing folder	Total
glioma	1,321	300	1,621
meningioma	1,339	306	1,645
notumor	1,595	405	2,000
pituitary	1,457	300	1,757
Total	**5,712**	**1,311**	**7,023**

Table E.3 Tensor shapes for ResNet152 (batch size B), as verified by the step-wise test scripts.

Stage	Shape	Notes
Input image	[B, 3, 224, 224]	RGB, resized, ImageNet-normalized
CNN feature map	[B, 2048, 7, 7]	512 channels for VGG, 1,536 for EfficientNet-B3
Spatial sequence → LSTM	[B, 49, 2048] → [B, 49, 256]	one layer, hidden 256, batch-first; last step kept: [B, 256]
Deep feature	[B, 128]	BatchNorm1d → Dropout(0.3) → Linear(256→128) → ReLU
Neural head	[B, 4]	Linear(128→4) logits; CrossEntropyLoss on raw logits
After scaler / PCA	[B, 128] / [B, 64]	both fitted on training features only

Environment. Ubuntu under WSL2, Python 3.12.14, torch 2.11.0+cu128, scikit-learn 1.9.0, NVIDIA RTX 5070 Laptop GPU (8 GB). Training used `drop_last=True` because

4,569 = 571 × 8 + 1 and BatchNorm1d cannot use a one-image batch. Every experimental number in this report is taken from the verified master deliverables (FINAL_PROJECT_REPORT, FINAL_PROJECT_EVIDENCE_MAP, FINAL_PROJECT_QC_REPORT: 174/174 automated checks, 75/75 hashes); repository paths are relative to the project root.

Table E.4 Principal scripts and records by study step.

Step	Script / record
Dataset check, split	`src/check_dataset.py` · `src/save_split.py` (seed 42, saved indices)
Augmentation	`src/data_pipeline_torch.py` L65–78
Step-wise verification	`test_resnet152_features.py`, `test_lstm_section.py`, `test_post_lstm_layers.py`, `test_complete_model.py`, `test_real_mri_batch.py`, `test_training_step.py`
Phase 1	`train_backbone.py` · `evaluate_backbone.py` · `evaluate_neural_head.py` · `diagnose_vgg16_collapse.py`
Leakage audit / policies	`final_project_qc.py` · `evaluate_leakage_clean_test.py` · post-audit record `FINAL_QC_BLOCKER.md`
Phase 2A / 2B	`train_backbone_30epoch.py` and Phase 2B run scripts under `phase2/`
Freeze	`frozen_15_manifest.json` (75 hashes) · `frozen_15_selection.csv`
Web-40	`audit_web_stress_40.py` · `lock_record.json` · `Web_Stress_Test_40_Source_List.xlsx`

Table E.5 Principal evidence sources by result family.

Result family	Repository source
Phase 1	`experiments/<backbone>/results/{final,neural_head,training}_results.json`
Leakage audit / policies	`final_project_docs/duplicate_leakage_audit/*.json` · `leakage_corrected_evaluation/leakage_corrected_summary.csv`
Phase 2A	`phase2/experiments_30epoch/<backbone>/…` · `phase2/results/leakage_corrected_30epoch/`
Phase 2B	`phase2/hyperparameter_dl/<backbone>/<run>/…`
Freeze	`phase2/external_validation/frozen_15_manifest.json`
BDNeuro	`bdneuro_overlap_audit/` · `bdneuro_exact_clean_cohort/` · `bdneuro_balanced_subset/` · `comparison_15_models_bdneuro_{full,balanced}/`
Web-40	`web_stress_40_audit/` · `web_stress_40_locked/lock_record.json` · `comparison_15_models_web_stress_40/`
XAI	`gradcam_evaluation/*.csv` · `occlusion_evaluation/*.csv`
Prototype	Final application: 15 frozen pipelines (5 backbones × Phase 1/2A/2B), Predict MRI and six saved-research tabs (Ch. 12). Final screenshots/specification document post-snapshot implementation state only, not accuracy evidence; supplied app.py is an earlier four-tab revision.
Historical items	VGG16 collapse (≈ 28%), legacy ResNet152 prototype (≈ 97.64%), NOR_01 pre-replacement audit: project records only (no JSON)

APPENDIX F
IMPLEMENTATION HISTORY AND GLOSSARY

This appendix preserves the implementation record that did not change the scientific design; the one event that did — the VGG collapse and frozen-VGG policy — is reported in Section 3.4. Fig. F.1 places the main decisions on the project timeline.

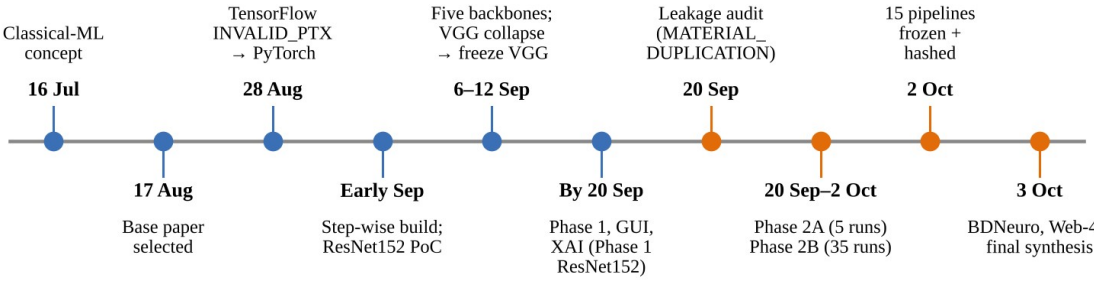


Fig. F.1 Project timeline.

Dates before 20 September come from project notes and are approximate; later dates come from repository output timestamps. Blue: build stage; orange: integrity and validation stage.

- **Classical concept (approximately July 2026).** A pipeline of median filtering, skull removal, K-means segmentation, handcrafted GLCM features and an RBF-SVM was briefly considered. Two corrections were recorded: a K-means cluster index cannot automatically be called "tumour", and dataset labels must not be relabelled benign or malignant. The direction was superseded around 17 August 2026 (approximate, from project notes), when a recent deep-learning base paper became a requirement.
- **TensorFlow failure and PyTorch migration (approximately 28 August 2026).** Because the base paper used TensorFlow, the first route was TensorFlow under WSL2. It detected the RTX 5070 GPU but realistic training failed with `INVALID_PTX`: the Blackwell architecture (compute capability sm_120) was unsupported by the build in use, and a nightly build proved unreliable. The project moved to PyTorch 2.11.0 + CUDA 12.8 and verified real GPU workloads before writing model code — an environmental decision, not a judgement about TensorFlow.
- **Step-wise verification and first proof of concept.** The model was built and checked stage by stage with executable scripts (Appendix E), which produced the tensor shapes above and settled that the LSTM reads spatial positions of one image. The first end-to-end hybrid was a root-level ResNet152 pipeline (RMSprop, LR 1e-4, best epoch 8); according to project notes its SVM reached about 97.64% test accuracy. It proved the complete path and was superseded.
- **Generalization to five backbones (approximately 6–12 September 2026).** The ResNet-specific code was replaced by a generic CNN–LSTM class with standardized artifact bundles under `experiments/<backbone>/`, one training

script and one evaluation script for all backbones. Fine-tuning VGG16 under this code produced the collapse described in Section 3.4.

Table F.1 Glossary of key terms.

Term	Meaning
Macro F1	Unweighted mean over classes of per-class F1; treats every class equally.
Neural head	Temporary Linear($128 \rightarrow 4$) classifier used to train the network; evaluated for comparison and used by Grad-CAM.
Duplicate leakage	Identical or near-identical images present in both training and evaluation partitions.
Material pair	Cross-split pair that is an exact file, exact decoded-pixel or high-confidence near-duplicate match.
material_clean	Post-hoc leakage-aware test subset with every image in a material pair removed (n = 1,061).
pHash	Perceptual hash of coarse image structure; a small Hamming distance means visually similar, not identical.
Exact-overlap-clean	External cohort after removing images pixel-identical to development images (BDNeuro: 2,644).
best_val_within_30 / epoch_30	Checkpoint with the highest neural validation accuracy within the 30-epoch run / the state after epoch 30.
Frozen pipeline	Checkpoint + scaler + PCA + SVM fixed and SHA-256 hashed before external performance evaluation.
Domain shift	Difference between the distribution of evaluation images and that of training images.